Topological Optics of Confined Light: The Fano-Week Compiler, the $\sqrt{13}$ Wall, the 16-Bit Vacuum Register, the Central Binary Edge, the Ogg Boundary, and the Algebraic Origin of Chemical Bonding

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Abstract

We give a strict algebraic–topological account of the periodic system, nuclear friction laws, chemical bonding, and the binary architecture of the vacuum register in the field

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}).$$

The central results are not numerical coincidences but theorems: radicals are binary opcodes, squares are compiled constants, multiplication of radicals is XOR in $\mathbb{F}_2^3$, the seven quadratic subfields of K form the Fano week, their discriminants multiply to

$$\Delta_K = 2^{16}3^4 5^4, \qquad \sqrt{|\Delta_K|} = 57\,600 = 240^2,$$

and the Pythagorean–Fibonacci cascade halts at the Kummer wall $\sqrt{13} \notin K$.

We further derive the field K as the unique rank-three maximal-constructibility squareclass field compatible with a three-cycle spatial torus and the Pythagorean–Fibonacci cascade. Moreover, we prove that no rank-three squareclass register can delay the first new prime beyond 13. Thus the wall $\sqrt{13} \notin K$ is not an arbitrary cutoff but the optimal and unavoidable overflow of a rank-three metric register.

The covolume is decoded as a static 16-bit holographic mask

$$57\,600 = \texttt{0xE100} = \texttt{1110 0001 0000 0000}_2,$$

whose four active bits encode the spatial valence dimensions $s + p$, whose twelve dormant bits encode $d + f$, and whose binary instruction set reproduces the capacity law, the spinor subshell suffix, the equatorial binary diode, the facing involution, the Krypton carry cascade, the FCC closure $172 = 43 \ll 2$, and the 137 register.

A deterministic register trace shows the Kummer wall as a 3-bit opcode overflow, the covolume as the compiled mask `0xE100`, the equatorial diode as a popcount identity, and the absolute closure $172 = 43 \ll 2$ as the first Ogg-breaking Heegner boundary. The facing involution is given a free topological model on $\mathbb{S}^4 \times \mathbb{T}^3$, whose register shadow is the byte-swap of the nuclear facing word.

We further identify a central binary edge of the FCC register. Its carry-free sum is the one-bit successor of the absolute closure and exposes the Kummer wall in the low nibble. The photon/antiphoton 1/0 assignment is interpreted as a crystal-domain edge encoding, not as standard QED charge conjugation; the facing map preserves parity and therefore supplies a geometric mirror, not a one-bit CP flip.

We prove the exact 2-adic formula for the binary Topological Friction Index, the equatorial binary diode, the Heegner congruence $p \equiv 19 \pmod{24}$ for $p > 11$, the O_h origin of valence angles, and the rigorous parity and Lichtenberg-compression cores of the Perez Hourglass. Conditional on explicit topological-optics postulates, chemical bonds are derived as phase-closed interference of confined gauge-light modes. Every non-arithmetic physical identification is explicitly separated as a conditional theorem, resonance, or open dynamical target.

We close by providing the first explicit numerical bridge from the toroidal spectrum to laboratory units: the lowest non-scalar kinetic quantum on T^3 yields

$$T_{\min} = \frac{\mathrm{Ry}}{3} \approx 4.535 \text{ eV},$$

which reproduces the H_2 dissociation energy $D_0 \approx 4.478$ eV to within 1.3% without free parameters. We further record five explicit falsification bets (absolute ceiling $Z = 172$, hydrogen bridge, equatorial alpha diode, technetium demotion, continuum compatibility) that distinguish the framework from post-hoc numerology.

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1 Epistemic protocol

To eliminate numerology, every statement is assigned one of four statuses.

Epistemic Status 1.1 (Level A: theorem). A statement proved from standard mathematics: algebraic number theory, representation theory, spectral geometry, coding theory, or elementary binary arithmetic.

Epistemic Status 1.2 (Level B: conditional theorem). A statement proved from Level A plus explicitly listed physical/geometric postulates of the IT^3 framework.

Epistemic Status 1.3 (Level C: resonance). A numerical or structural coincidence that is not used as a proof and does not enter predictions unless separately dynamized.

Epistemic Status 1.4 (Level O: open target). A precisely formulated dynamical problem whose solution would derive a Level B postulate as a theorem. Level O statements are not used as evidence and carry no numerical authority.

Example. The identity

$$13^2 - 32 = 137$$

is Level A. The statement that 137 is the physical inverse fine-structure constant is Level C unless a separate dynamical derivation is supplied. The statement that the Kummer wall follows from a spectral-action minimum is Level O until proved.

2 The squareclass compiler: roots as opcodes

Let

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}).$$

For $n \in \mathbb{N}$, write

$$n = a^2 d,$$

where d is squarefree. Then

$$\sqrt{n} = a\sqrt{d}.$$

Lemma 2.1 (Quadratic subfields of K)**.** *The Galois group is*

$$\mathrm{Gal}(K/\mathbb{Q}) \simeq (\mathbb{Z}/2)^3.$$

Hence K has exactly $2^3 - 1 = 7$ quadratic subfields, namely

$$\mathbb{Q}(\sqrt{2}), \quad \mathbb{Q}(\sqrt{3}), \quad \mathbb{Q}(\sqrt{5}), \quad \mathbb{Q}(\sqrt{6}), \quad \mathbb{Q}(\sqrt{10}), \quad \mathbb{Q}(\sqrt{15}), \quad \mathbb{Q}(\sqrt{30}).$$

Proof. The extensions $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{5})$ are independent quadratic extensions, so

$$[K : \mathbb{Q}] = 2^3 = 8, \qquad \mathrm{Gal}(K/\mathbb{Q}) \simeq (\mathbb{Z}/2)^3.$$

Quadratic subfields correspond to index-2 subgroups, equivalently to nonzero characters

$$\chi : \mathrm{Gal}(K/\mathbb{Q}) \to \{\pm 1\}.$$

There are $2^3 - 1 = 7$ such characters. The fixed fields are generated by square roots of all nonempty squarefree products of $2, 3, 5$, giving the listed seven fields. This is the standard Kummer description of multiquadratic extensions [3]. □

Theorem 2.2 (Squareclass selection rule)**.**

$$\sqrt{n} \in K \iff \mathrm{sf}(n) \in \{1, 2, 3, 5, 6, 10, 15, 30\}.$$

Equivalently,

$$\sqrt{n} \in K \iff \mathrm{sf}(n) \mid 30.$$

Proof. Since $a \in \mathbb{Q}$, $\sqrt{n} \in K$ iff $\sqrt{d} \in K$. By Lemma 2.1, the quadratic subfields of K are exactly the seven fields listed there. Therefore $\sqrt{d} \in K$ iff d is one of

$$1, 2, 3, 5, 6, 10, 15, 30.$$

These are precisely the squarefree integers whose prime divisors belong to $\{2, 3, 5\}$, i.e. divide 30. □

Definition 2.3 (Binary opcode of a radical)**.** For squarefree

$$d = 2^{b_2} 3^{b_3} 5^{b_5}, \qquad b_p \in \{0, 1\},$$

define

$$\mathrm{op}(d) = (b_5, b_3, b_2) \in \mathbb{F}_2^3.$$

d	opcode	ordinary binary	role
1	000	1	compiled scalar unit
2	001	10	spinor bit
3	010	11	color bit
5	100	101	curvature/f-metric bit
6	011	110	$2 \oplus 3$
10	101	1010	$2 \oplus 5$
15	110	1111	$3 \oplus 5$
30	111	11110	full register saturation

Proposition 2.4 (XOR arithmetic of radicals)**.** *If d_1, d_2 are squarefree and $\mathrm{sf}(d_1 d_2)$ denotes the squarefree part of $d_1 d_2$, then*

$$\mathrm{op}(\mathrm{sf}(d_1 d_2)) = \mathrm{op}(d_1) \oplus \mathrm{op}(d_2).$$

Proof. In $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$, multiplication corresponds to addition of exponent vectors modulo 2. Removing square factors is exactly reduction modulo 2 in each prime coordinate. Hence the opcode adds by XOR. □

Proposition 2.5 (Squaring is compilation)**.** *For squarefree d,*

$$(\sqrt{d})^2 = d \rightsquigarrow \mathrm{op} = 000$$

in the squareclass register.

Proof. The square d is a rational number, hence its squareclass is trivial. In the opcode language the active radical bits are erased. □

Example 2.6 (The number 8 is compiled, not new)**.** *Since*

$$8 = 2^2 \cdot 2,$$

we have

$$\sqrt{8} = 2\sqrt{2}.$$

Thus the ordinary binary of 8 *is* `1000`*, but the squareclass opcode is again* `001`*. The radical content is already an old bit with a compiled square multiplier.*

3 Derivation of the metric field: why $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$?

The choice

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$$

is not an arbitrary assignment of primes. It is forced by the topological rank of the spatial torus, the additive composition of orthogonal metric tensions, and the requirement that the vacuum construct as much of the Pythagorean–Fibonacci cascade as possible before introducing a new independent squareclass.

Let

$$\Gamma \subset \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$$

be the squareclass group of allowed metric tensions.

Postulate 3.1 (Toroidal metric rank)**.** The spatial vacuum fibre is T^3, and the metric squareclass group is dual to its mod-2 homology:

$$\Gamma \simeq H^1(T^3; \mathbb{F}_2).$$

Hence

$$\dim_{\mathbb{F}_2} \Gamma = 3.$$

Postulate 3.2 (Orthogonal composition)**.** If two allowed metric tensions have squared lengths a and b and are orthogonal, then the resultant squared length $a + b$ is also allowed. Therefore the cascade of primitive squared tensions is generated by additive composition.

Postulate 3.3 (Pythagorean–Fibonacci cascade)**.** The primitive squared metric tensions begin

$$x_1 = 1, \qquad x_2 = 2, \qquad x_n = x_{n-1} + x_{n-2}.$$

Equivalently,

$$x_n = F_{n+1}.$$

The limiting ratio is the golden ratio:

$$\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = \varphi, \qquad \varphi^2 = \varphi + 1.$$

Postulate 3.4 (Maximal constructible initial segment)**.** Among all rank-three squareclass groups, Γ is chosen to contain the longest initial segment of the cascade squareclasses $\mathrm{sf}(x_n)$. In other words, the vacuum does not introduce a new independent prime metric direction earlier than necessary.

Theorem 3.5 (Uniqueness of the metric field)**.** *Under Postulates 3.1–3.4, the only possible squareclass group is*

$$\Gamma = \langle 2, 3, 5 \rangle_{\mathbb{F}_2} = \{1, 2, 3, 5, 6, 10, 15, 30\},$$

and therefore

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}).$$

Moreover,

$$\sqrt{x_n} \in K \quad \textit{for} \quad 1 \le n \le 5,$$

but

$$\sqrt{x_6} = \sqrt{13} \notin K.$$

Thus the Kummer wall is the first necessary rank-three overflow of the metric register.

Proof. The first cascade terms are

$$x_1 = 1, \quad x_2 = 2, \quad x_3 = 3, \quad x_4 = 5, \quad x_5 = 8, \quad x_6 = 13.$$

Their squarefree parts are

$$\mathrm{sf}(x_1) = 1, \quad \mathrm{sf}(x_2) = 2, \quad \mathrm{sf}(x_3) = 3, \quad \mathrm{sf}(x_4) = 5, \quad \mathrm{sf}(x_5) = 2, \quad \mathrm{sf}(x_6) = 13.$$

To admit $x_2 = 2$, the group Γ must contain the squareclass of 2. To admit $x_3 = 3$, it must contain the squareclass of 3. To admit $x_4 = 5$, it must contain the squareclass of 5. The classes of $2, 3, 5$ are independent in $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$, because they are distinct primes. Hence they already span a three-dimensional $\mathbb{F}_2$-subspace.

By Postulate 3.1,

$$\dim_{\mathbb{F}_2} \Gamma = 3.$$

Therefore no further independent prime class can be present, and

$$\Gamma = \langle 2, 3, 5 \rangle_{\mathbb{F}_2}.$$

This is precisely the squareclass group of

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}).$$

Now

$$x_5 = 8 = 2^2 \cdot 2,$$

so

$$\mathrm{sf}(x_5) = 2 \in \Gamma,$$

and therefore

$$\sqrt{8} = 2\sqrt{2} \in K.$$

But

$$x_6 = 13$$

is prime and

$$13 \notin \{2, 3, 5\}.$$

Hence

$$\mathrm{sf}(x_6) = 13 \notin \Gamma,$$

so

$$\sqrt{13} \notin K.$$

Finally, the maximal-delay principle forces this particular rank-three field. Any other rank-three field omitting one of $2, 3, 5$ stops the cascade earlier. For example, $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{7})$ fails already at $x_4 = 5$. Thus $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ is the unique rank-three field that constructs the maximal initial segment

$$1, 2, 3, 5, 8$$

before the first unavoidable new prime 13 appears. □

Theorem 3.6 (Optimal rank-three cascade delay)**.** *Let $\Gamma \subset \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ be any squareclass group of $\mathbb{F}_2$-dimension 3. Then the longest initial segment of cascade squareclasses*

$$\mathrm{sf}(x_1), \mathrm{sf}(x_2), \ldots$$

contained in Γ has length at most 5. Equality holds if and only if

$$\Gamma = \langle 2, 3, 5 \rangle_{\mathbb{F}_2}.$$

Consequently, no rank-three metric register can delay the first new independent squareclass beyond 13.

Proof. The first six cascade squarefree parts are

$$1, 2, 3, 5, 2, 13.$$

To contain the first five nontrivial requirements, a group must contain the classes of 2, 3, and 5. These three classes are independent, so any rank-three group containing them is exactly their span:

$$\Gamma = \langle 2, 3, 5 \rangle_{\mathbb{F}_2}.$$

To also contain 13, the group would have to contain four independent classes $2, 3, 5, 13$, forcing rank at least 4. Hence rank three cannot pass $x_6 = 13$. The unique rank-three group achieving length five is $\langle 2, 3, 5 \rangle$. □

Corollary 3.7 (The wall is optimal, not chosen)**.** *The Kummer wall*

$$\sqrt{13} \notin K$$

is not a convenient cutoff inserted after the fact. It is the unavoidable first overflow of any rank-three squareclass register trying to construct the Pythagorean–Fibonacci cascade.

rank-three squareclass group	first cascade squareclass missed	stopping index
$\langle 2, 3, 5 \rangle$	13	x_6
$\langle 2, 3, 7 \rangle$	5	x_4
$\langle 2, 5, 7 \rangle$	3	x_3
$\langle 3, 5, 7 \rangle$	2	x_2

Numerology Guard 3.8 (Status of the derivation). Theorems 3.5 and 3.6 are Level A conditional on the explicitly stated structural postulates. They remove the arbitrariness of choosing $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ among multiquadratic fields and show that the wall at 13 is optimal for rank three. They do not yet derive the cascade itself from the Standard Model; that remains the main dynamical open problem, formulated as Level O targets in Appendix G.

4 The Fano week and the holographic covolume

Theorem 4.1 (Seven days as seven nonzero states)**.** *The seven quadratic subfields of K are exactly the seven nonzero vectors of $\mathbb{F}_2^3$.*

Proof. This is Lemma 2.1 reformulated in opcode language: the nonzero vectors of $\mathbb{F}_2^3$ are

$$\texttt{001}, \texttt{010}, \texttt{011}, \texttt{100}, \texttt{101}, \texttt{110}, \texttt{111},$$

corresponding to

$$2, 3, 6, 5, 10, 15, 30.$$

□

Theorem 4.2 (Discriminant product)**.** *The discriminants of the seven quadratic subfields are*

$$8,\ 12,\ 5,\ 24,\ 40,\ 60,\ 120.$$

Their product is

$$\Delta_K = 8 \cdot 12 \cdot 5 \cdot 24 \cdot 40 \cdot 60 \cdot 120 = 2^{16}3^4 5^4 = 3\,317\,760\,000.$$

Proof. For a squarefree integer d, the discriminant of $\mathbb{Q}(\sqrt{d})$ is

$$\Delta_{\mathbb{Q}(\sqrt{d})} = \begin{cases} d, & d \equiv 1 \pmod 4, \\ 4d, & d \equiv 2, 3 \pmod 4. \end{cases}$$

Applying this to $d = 2, 3, 5, 6, 10, 15, 30$ gives

$$8, 12, 5, 24, 40, 60, 120.$$

Since $K/\mathbb{Q}$ is abelian, the Führer–Diskriminanten–Satz gives

$$\Delta_K = \prod_{\chi \neq 1} f_\chi,$$

where the product runs over the seven nontrivial quadratic characters. In this case the conductors are exactly the seven discriminants above. Prime factorization gives

$$8 = 2^3, \quad 12 = 2^2 \cdot 3, \quad 5 = 5, \quad 24 = 2^3 \cdot 3, \quad 40 = 2^3 \cdot 5, \quad 60 = 2^2 \cdot 3 \cdot 5, \quad 120 = 2^3 \cdot 3 \cdot 5.$$

Summing exponents:

$$2:\ 3+2+0+3+3+2+3 = 16,$$
$$3:\ 0+1+0+1+0+1+1 = 4,$$
$$5:\ 0+0+1+0+1+1+1 = 4.$$

Hence

$$\Delta_K = 2^{16}3^4 5^4.$$

□

Theorem 4.3 (Covolume as the square of the E_8 kissing number)**.** *Since K is totally real, the Minkowski covolume of $\mathcal{O}_K$ is*

$$\sqrt{|\Delta_K|} = 2^8 3^2 5^2 = 57\,600 = 240^2.$$

Here 240 is the kissing number of the E_8 lattice.

Proof. For a totally real number field of degree 8, the covolume of the ring of integers in the Minkowski space $\mathbb{R}^8$ is $\sqrt{|\Delta_K|}$. Using Theorem 4.2,

$$\sqrt{\Delta_K} = \sqrt{2^{16}3^4 5^4} = 2^8 3^2 5^2.$$

But

$$2^8 3^2 5^2 = (2^4 \cdot 3 \cdot 5)^2 = 240^2.$$

The equality $\tau_8 = 240$ for the E_8 root lattice is a classical theorem of sphere packing [6]. □

Theorem 4.4 (The 16-bit holographic mask)**.**

$$57\,600 = 1110\ 0001\ 0000\ 0000_2 = \texttt{0xE100}.$$

Its Hamming weight is

$$\mathrm{wt}(57\,600) = 4.$$

Proof. We have

$$57\,600 = 225 \cdot 256.$$

Now

$$225 = 15^2 = 1110\ 0001_2,$$

and multiplication by $256 = 2^8$ is a left shift by eight bits. Therefore

$$57\,600 = 1110\ 0001\ 0000\ 0000_2.$$

Counting ones gives $3 + 1 = 4$. □

Remark 4.5 (O_h interpretation of the mask). The O_h spatial dimensions before spinor doubling are

$$\dim s + \dim p + \dim d + \dim f = 1 + 3 + 5 + 7 = 16.$$

The four active bits of the mask correspond to the active valence dimensions

$$s + p = 1 + 3 = 4,$$

while the twelve zero bits correspond to the dormant dimensions

$$d + f = 5 + 7 = 12.$$

After spinor doubling the maximal floor capacity is

$$16 \cdot 2 = 32.$$

5 The Kummer wall and the hidden compiled twelve

The cascade introduced in Section 3 is

$$x_1 = 1, \qquad x_2 = 2, \qquad x_n = x_{n-1} + x_{n-2}.$$

Equivalently,

$$x_n = F_{n+1},$$

where F_n is the Fibonacci sequence.

Theorem 5.1 (Cascade halt at 13)**.** *The first forbidden cascade term is*

$$x_6 = 13,$$

because

$$\sqrt{13} \notin K.$$

Proof. The first terms are

$$1, 2, 3, 5, 8, 13, 21, 34, \ldots$$

Their squarefree parts are

$$1, 2, 3, 5, 2, 13, 21, 34, \ldots$$

By Theorem 2.2, allowed squarefree parts are exactly divisors of 30 composed of primes $2, 3, 5$. The terms

$$1, 2, 3, 5, 2$$

are allowed. But 13 is prime and

$$13 \nmid 30,$$

so

$$\sqrt{13} \notin K.$$

Thus the constructible cascade stops at $x_6 = 13$. □

Proposition 5.2 (The wall is selective, not a squareclass ceiling)**.** *One has*

$$\sqrt{15} \in K, \qquad \sqrt{30} \in K,$$

because

$$15 \mid 30, \qquad 30 \mid 30.$$

Nevertheless,

$$\sqrt{13} \notin K.$$

Hence the wall at 13 *is not a squareclass ceiling. It is the first overflow of the additive Pythagorean–Fibonacci cascade*

$$1,\ 2,\ 3,\ 5,\ 8,\ 13, \ldots$$

inside a rank-three squareclass register.

Proof. By Theorem 2.2,

$$\sqrt{n} \in K \iff \mathrm{sf}(n) \mid 30.$$

Since $15 \mid 30$ and $30 \mid 30$, both $\sqrt{15}$ and $\sqrt{30}$ belong to K. But 13 is prime and $13 \nmid 30$, so $\sqrt{13} \notin K$. □

Numerology Guard 5.3 (Correct wording). Use the phrases

"additive cascade wall" or "rank-three constructibility overflow".

Do not write an unconditional "UV continuum is forbidden by $\sqrt{13} \notin K$" unless one separately postulates that all propagating metric modes are generated by the additive cascade. Otherwise the statement is false, because $\sqrt{15}$ and $\sqrt{30}$ are algebraically allowed.

Corollary 5.4 (Product of allowed fruits)**.** *The product of the allowed cascade terms before the wall is*

$$1 \cdot 2 \cdot 3 \cdot 5 \cdot 8 = 240.$$

Proof. Direct arithmetic. □

Corollary 5.5 (Additive shadows of the wall)**.** *One has*

$$1 + 2 + 3 + 5 + 8 + 13 = 32,$$

and

$$1 + 2 + 3 + 5 + 8 + 13 + 21 + 34 = 87.$$

Proof. Direct arithmetic. The first sum is the equatorial capacity $C(5) = 32$. The second is $86 + 1$, the ionization-limit register. □

Theorem 5.6 (Cohn's Fibonacci-square theorem)**.** *The only squares in the Fibonacci sequence are*

$$F_0 = 0, \qquad F_1 = F_2 = 1, \qquad F_{12} = 144 = 12^2.$$

This is a deep theorem of J. H. E. Cohn [2]. In the cascade notation,

$$x_{11} = F_{12} = 144 = 12^2.$$

Conditional Theorem 5.7 (Unique Fibonacci-square global scale)**.** Assume the following architectural axiom.

Axiom 5.8 (FS)**.** The global number of floors L must satisfy

$$L^2 = x_n$$

for some cascade term x_n, with $L > 1$.

Then

$$L = 12.$$

Proof. By Cohn's theorem, the only nontrivial square cascade term is

$$x_{11} = 144 = 12^2.$$

Hence $L^2 = 144$, so $L = 12$. □

Numerology Guard 5.9. Cohn's theorem does not by itself prove that the physical periodic system has twelve floors. It proves that *if* the architecture requires a nontrivial Fibonacci-square compiled scale, then that scale is uniquely 12. The modular normalization

$$1728 = 12^3$$

is then an exact arithmetic shadow of the same compiled scale.

6 Heegner primes and the rigorous step 24

Theorem 6.1 (Heegner congruence modulo 24)**.** *Let $p > 11$ be a prime such that*

$$\mathbb{Q}(\sqrt{-p})$$

has class number one. Then

$$p \equiv 19 \pmod{24}.$$

Consequently, for any two such primes p, q,

$$24 \mid (p - q).$$

Proof. A positive definite primitive binary quadratic form is written

$$(a, b, c): \quad ax^2 + bxy + cy^2,$$

with discriminant

$$D = b^2 - 4ac < 0.$$

The class number of an imaginary quadratic field equals the number of reduced primitive forms of the corresponding fundamental discriminant. A reduced form satisfies

$$|b| \le a \le c,$$

with the usual sign conventions when equality holds.

First suppose $p \equiv 1 \pmod 4$. Then the fundamental discriminant is $-4p$. Consider the form

$$\left(2, 2, \frac{p+1}{2}\right).$$

Its discriminant is

$$2^2 - 4 \cdot 2 \cdot \frac{p+1}{2} = 4 - 4(p+1) = -4p.$$

Since $p = 4k+1$, the third coefficient is $2k+1$, odd, so the form is primitive. It is reduced because

$$|2| = 2 \leq 2 \leq \frac{p+1}{2}$$

for $p \geq 5$, and it is nonprincipal because $a = 2 > 1$. Hence the class number is greater than one, contradiction. Therefore a Heegner prime $p > 11$ must satisfy

$$p \equiv 3 \pmod 4.$$

Now assume

$$p \equiv 7 \pmod 8, \qquad p > 11.$$

Then $p \geq 23$, and

$$c = \frac{p+1}{8}$$

is an integer with $c \geq 3$. The form

$$(2, 1, c)$$

has discriminant

$$1^2 - 4 \cdot 2 \cdot c = 1 - 8 \cdot \frac{p+1}{8} = -p.$$

It is primitive and reduced because

$$1 \leq 2 \leq c.$$

It is nonprincipal because $a = 2 > 1$. Contradiction. Hence

$$p \not\equiv 7 \pmod 8.$$

This excludes the residues

$$p \equiv 7, 23 \pmod{24}.$$

Finally assume

$$p \equiv 11 \pmod{12}, \qquad p > 11.$$

The case $p = 23$ has already been excluded because $23 \equiv 7 \pmod 8$. Thus $p \geq 47$, and

$$c = \frac{p+1}{12} \geq 4.$$

The form

$$(3, 1, c)$$

has discriminant

$$1^2 - 4 \cdot 3 \cdot c = 1 - 12 \cdot \frac{p+1}{12} = -p.$$

It is primitive and reduced because

$$1 \leq 3 \leq c.$$

It is nonprincipal because $a = 3 > 1$. Contradiction. Hence

$$p \not\equiv 11 \pmod{12}.$$

This excludes the residue

$$p \equiv 11 \pmod{24},$$

and also 23, already excluded.

For a prime $p > 3$ with $p \equiv 3 \pmod 4$, the possible residues modulo 24 are

$$7,\ 11,\ 19,\ 23.$$

We excluded $7, 11, 23$. Therefore

$$p \equiv 19 \pmod{24}.$$

□

Corollary 6.2 (Step 24)**.** *For Heegner primes $p, q > 11$,*

$$24 \mid (p - q).$$

In particular,

$$43 - 19 = 24, \qquad 67 - 43 = 24, \qquad 163 - 67 = 96 = 4 \cdot 24.$$

Remark 6.3 (Role of the eta function). The identity

$$\Delta(\tau) = (2\pi)^{12}\eta(\tau)^{24}$$

explains why 24 is natural in modular theory. But the Heegner step itself is proved above by quadratic forms, not by phase mysticism [5, 8, 9].

7 Binary diode: exact 2-adic friction law

For an alpha-type transition

$$Z \to Z - 2,$$

define the binary Topological Friction Index by

$$\mathrm{TFI}(Z \to Z - 2) = \mathrm{wt}\big(Z \oplus (Z - 2)\big),$$

where $\oplus$ is bitwise exclusive OR and wt is Hamming weight.

Lemma 7.1 (Basic borrow identity)**.** *For $m \geq 1$,*

$$\mathrm{wt}\big(m \oplus (m - 1)\big) = \nu_2(m) + 1.$$

Proof. Write

$$m = 2^t u,$$

with u odd. Then the binary expansion of m ends with exactly t zeros preceded by a 1. Subtracting 1 turns that 1 into 0 and the trailing t zeros into ones. Higher bits are unchanged. Therefore $m \oplus (m-1)$ has exactly $t + 1$ low-order ones and no other ones. Hence its weight is $t + 1 = \nu_2(m) + 1$. □

Theorem 7.2 (Exact 2-adic formula for TFI)**.** *For $Z \geq 2$,*

$$\mathrm{TFI}(Z \to Z - 2) = \begin{cases} \nu_2(Z), & Z \text{ even}, \\ \nu_2(Z - 1), & Z \text{ odd}. \end{cases}$$

Proof. If Z is even, write $Z = 2m$. Then

$$Z - 2 = 2(m - 1).$$

Bitwise left shift commutes with XOR:

$$Z \oplus (Z - 2) = 2m \oplus 2(m - 1) = 2\big(m \oplus (m - 1)\big).$$

Shifting left does not change Hamming weight, so by Lemma 7.1

$$\mathrm{TFI} = \mathrm{wt}\big(m \oplus (m - 1)\big) = \nu_2(m) + 1 = \nu_2(2m) = \nu_2(Z).$$

If Z is odd, write $Z = 2m + 1$ with $m \geq 1$. Then

$$Z - 2 = 2m - 1 = 2(m - 1) + 1.$$

The least significant bits of Z and $Z - 2$ are both 1, so they cancel in XOR. Hence

$$Z \oplus (Z - 2) = 2\big(m \oplus (m - 1)\big).$$

Therefore

$$\mathrm{TFI} = \mathrm{wt}\big(m \oplus (m - 1)\big) = \nu_2(m) + 1 = \nu_2(2m) = \nu_2(Z - 1).$$

□

Theorem 7.3 (Equatorial binary diode)**.** *Let E be an even closure such that*

$$E \equiv 6 \pmod 8.$$

Then the even rail

$$E + 2 \to E$$

has

$$\mathrm{TFI} \geq 3,$$

while the odd rail

$$E + 1 \to E - 1$$

has

$$\mathrm{TFI} = 1.$$

Proof. Since $E \equiv 6 \pmod 8$, we have

$$\nu_2(E) = 1.$$

Also

$$E + 2 \equiv 0 \pmod 8,$$

so

$$\nu_2(E + 2) \geq 3.$$

By Theorem 7.2,

$$\mathrm{TFI}(E + 2 \to E) = \nu_2(E + 2) \geq 3.$$

For the odd rail,

$$\mathrm{TFI}(E + 1 \to E - 1) = \nu_2((E + 1) - 1) = \nu_2(E) = 1.$$

□

Example 7.4 (Relativistic equator $Z = 86$)**.**

$$86 = 0101\ 0110_2 \equiv 6 \pmod 8.$$

Therefore

$$88 \to 86 : \quad \mathrm{TFI} = \nu_2(88) = 3,$$

because $88 = 8 \cdot 11$, *while*

$$87 \to 85 : \quad \mathrm{TFI} = \nu_2(86) = 1.$$

The diode is pure binary arithmetic.

8 ATFI: continuous toroidal metric friction

The spatial torus is generated by orthogonal metric roots

$$1, \quad \sqrt{2}, \quad \sqrt{3}.$$

The f-block corresponds to $\sqrt{5}$, orthogonal to the spatial T^3.

For changes in block occupations $\Delta n_s, \Delta n_p, \Delta n_d$, define

$$\mathrm{ATFI}^2 = (\Delta n_s)^2 + 2(\Delta n_p)^2 + 3(\Delta n_d)^2.$$

Proposition 8.1 (ATFI as Euclidean norm on T^3)**.** *The quantity* ATFI *is exactly the Euclidean length of the jump vector*

$$\Delta\vec{v} = \Delta n_s \cdot 1 + \Delta n_p \cdot \sqrt{2} + \Delta n_d \cdot \sqrt{3}$$

in the orthogonal metric generated by $1, \sqrt{2}, \sqrt{3}$.

Proof. Because the generators are orthogonal,

$$\|\Delta\vec{v}\|^2 = (\Delta n_s)^2\|1\|^2 + (\Delta n_p)^2\|\sqrt{2}\|^2 + (\Delta n_d)^2\|\sqrt{3}\|^2 = (\Delta n_s)^2 + 2(\Delta n_p)^2 + 3(\Delta n_d)^2.$$

□

transition type	ATFI	interpretation
pure f-block, $\Delta n_f \neq 0$	0	no spatial projection
pure s-block, -2	2	polar-axis jump
pure p-block, -2	$\sqrt{8} \approx 2.828$	planar lattice jump
pure d-block, -2	$\sqrt{12} \approx 3.464$	volumetric overtone jump
$s+p$, each -1	$\sqrt{3} \approx 1.732$	mixed equatorial slide

Thus

$$\mathrm{TFI} = \text{discrete borrow cascade}, \qquad \mathrm{ATFI} = \text{continuous toroidal metric friction}.$$

9 The Perez Hourglass: rigorous parity and Lichtenberg compression

The Perez Hourglass is a bidirectional fractal extension of Pascal's triangle [10]. To avoid unproved metaphysical leaps, we isolate its rigorous mathematical core: a subtractive Pascal triangle, parity preservation modulo 2, row balance, and the Lichtenberg compression identity.

Definition 9.1 (Subtractive Pascal triangle)**.** Define integer rows $S(n,k)$ for $n \geq 0$, $0 \leq k \leq n$, by

$$S(0,0) = 1,$$

and for $n \geq 1$,

$$S(n,0) = S(n,n) = 1, \qquad S(n,k) = S(n-1,k-1) - S(n-1,k) \quad (1 \leq k \leq n-1).$$

The first rows are

$$\begin{aligned}
S(0) &= (1),\\
S(1) &= (1,1),\\
S(2) &= (1,0,1),\\
S(3) &= (1,1,-1,1),\\
S(4) &= (1,0,2,-2,1),\\
S(5) &= (1,1,-2,4,-3,1),\\
S(6) &= (1,0,3,-6,7,-4,1),\\
S(7) &= (1,1,-3,9,-13,11,-5,1).
\end{aligned}$$

Theorem 9.2 (Parity preservation: rigorous core of Hourglass Theorem 7)**.** *For all $n \geq 0$ and $0 \leq k \leq n$,*

$$S(n,k) \equiv \binom{n}{k} \pmod 2.$$

Consequently, the modulo-2 bitmap of the subtractive hemisphere is exactly the Sierpinski gasket.

Proof. The binomial coefficients satisfy Pascal's recurrence

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

with boundary values $\binom{n}{0} = \binom{n}{n} = 1$. Modulo 2, subtraction and addition are identical because

$$-1 \equiv 1 \pmod 2.$$

Thus $S(n,k) \bmod 2$ satisfies the same recurrence and the same boundary conditions as $\binom{n}{k} \bmod 2$. Induction on n gives

$$S(n,k) \equiv \binom{n}{k} \pmod 2.$$

The classical theorem that Pascal's triangle modulo 2 generates the Sierpinski gasket then applies verbatim to the subtractive triangle. □

Proposition 9.3 (Perfect row balance)**.** *For every $n \geq 1$,*

$$\sum_{k=0}^{n} S(n,k) = 2.$$

Equivalently, the signed sum of the interior entries is exactly 0.

Proof. For $n \geq 2$, using the interior recurrence,

$$\sum_{k=1}^{n-1} S(n,k) = \sum_{k=1}^{n-1} \big(S(n-1,k-1) - S(n-1,k)\big).$$

This telescopes:

$$\sum_{k=1}^{n-1} S(n,k) = S(n-1,0) - S(n-1,n-1) = 1 - 1 = 0.$$

Adding the two boundary entries $S(n,0) = S(n,n) = 1$ gives total row sum 2. For $n = 1$ the statement is immediate. □

Definition 9.4 (Lichtenberg numbers)**.** For $n \geq 1$, define

$$L_n = \left\lfloor \frac{2^n}{3} \right\rfloor.$$

The first values are

$$L_1 = 0, \quad L_2 = 1, \quad L_3 = 2, \quad L_4 = 5, \quad L_5 = 10, \quad L_6 = 21, \quad L_7 = 42, \quad L_8 = 85.$$

Equivalently,

$$L_n = \frac{2^n - \varepsilon_n}{3}, \qquad \varepsilon_n = \begin{cases} 2, & n \text{ odd}, \\ 1, & n \text{ even}. \end{cases}$$

Theorem 9.5 (Lichtenberg compression identity: rigorous core of Hourglass Theorem 8)**.** *For every* $n \geq 1$*,*

$$L_{n+2} = 2^n + L_n.$$

In particular,

$$L_8 = 2^6 + L_6 = 64 + 21 = 85.$$

Proof. Since $2^n \bmod 3$ equals 2 for odd n and 1 for even n, the floor formula gives

$$L_n = \frac{2^n - \varepsilon_n}{3},$$

with $\varepsilon_{n+2} = \varepsilon_n$. Therefore

$$L_{n+2} = \frac{2^{n+2} - \varepsilon_{n+2}}{3} = \frac{4 \cdot 2^n - \varepsilon_n}{3}.$$

But

$$2^n + L_n = 2^n + \frac{2^n - \varepsilon_n}{3} = \frac{3 \cdot 2^n + 2^n - \varepsilon_n}{3} = \frac{4 \cdot 2^n - \varepsilon_n}{3}.$$

Hence

$$L_{n+2} = 2^n + L_n.$$

□

Numerology Guard 9.6 (Epistemic guard: biology and quantum memory)*.* The parity theorem and the Lichtenberg compression identity are Level A. Their interpretations as

- compression of 64 codons into 21 functional meanings,
- double-strand DNA $128 = 2^7$ states,
- Perez Hourglass Associative Memory (PHAM),
- fractal quantum error correction,

are Level C resonances or architectural directions unless a separate biochemical or quantum-informational dynamical model is supplied.

10 Topological optics of the vacuum

Let the spatial vacuum fibre be

$$T^3 = \mathbb{R}^3/\Lambda,$$

where Λ is generated by orthogonal roots

$$1, \quad \sqrt{2}, \quad \sqrt{3}.$$

Theorem 10.1 (Discrete photon wavevectors)**.** *Allowed photon wavevectors belong to the dual lattice* Λ^**, and their squared norms are*

$$\|\vec{k}\|^2 = (2\pi)^2 \left(n_1^2 + \frac{n_2^2}{2} + \frac{n_3^2}{3} \right), \qquad n_i \in \mathbb{Z}.$$

Proof. A plane wave $e^{i\vec{k}\cdot\vec{x}}$ is single-valued on $T^3 = \mathbb{R}^3/\Lambda$ iff

$$\vec{k} \cdot \lambda \in 2\pi\mathbb{Z}$$

for every $\lambda \in \Lambda$. Hence $\vec{k} \in \Lambda^*$. For the orthogonal basis of lengths $1, \sqrt{2}, \sqrt{3}$, the dual basis has lengths 2π, $2\pi/\sqrt{2}$, $2\pi/\sqrt{3}$. Therefore an integral dual vector has squared norm as stated. □

Postulate 10.2 (Constructible cascade metric tension)**.** Every propagating vacuum mode that is generated by the additive Pythagorean–Fibonacci cascade must have metric tension belonging to $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$.

Theorem 10.3 (Additive cascade wall, not squareclass ceiling)**.** *Under Postulate 10.2, the vacuum admits no cascade mode requiring metric tension* $\sqrt{13}$*, because*

$$\sqrt{13} \notin K.$$

However, this is not *a statement that all metric tensions above* $\sqrt{13}$ *are forbidden. The field* K *contains* $\sqrt{15}$ *and* $\sqrt{30}$*, both larger than* $\sqrt{13}$*. The wall at* 13 *is specifically the first overflow of the* additive cascade *inside a rank-three squareclass register.*

Proof. This is Theorem 5.1 combined with Postulate 10.2. The cascade of allowed squared tensions stops at 13. A cascade mode requiring $\sqrt{13}$ would demand a new prime outside the squareclass alphabet $\{2, 3, 5\}$. By Proposition 5.2, $\sqrt{15} \in K$ and $\sqrt{30} \in K$, so the obstruction is cascade-specific, not a global UV bound on the field. □

Conditional Theorem 10.4 (Mass as confined light)**.** Assume the topological-optics postulate:

Axiom 10.5 (Confined-light ontology)**.** Fermionic matter is gauge light undergoing total internal topological reflection at the Kummer wall, forming a standing wave whose localized energy is rest mass.

Then a high-energy gauge fluctuation reaching the algebraic boundary has group velocity

$$v_g \to 0,$$

reflects as its CP-conjugate mirror mode, interferes with the incident wave, and produces a localized standing-wave energy packet. This packet is a fermionic mass state.

Logical derivation under the axiom. By the cascade-wall theorem, the mode cannot propagate beyond the algebraic boundary. Total reflection produces an incident and a reflected wave. Their superposition is stationary. Stationary localized energy is, by definition, rest mass in the confined-light ontology. □

Numerology Guard 10.6. This is Level B, not Level A. The algebraic wall is theorem; the identification of mass with confined light is an explicit ontological postulate. Its dynamical derivation is Level O target 4 in Appendix G.

11 Chemistry as interference of confined light

We now enter the conditional but logically coherent chemical layer.

Postulate 11.1 (Chemical optics axioms)**.** C1: Atoms are topological optical resonators labelled by states $|S_n, \Lambda, \mu\rangle$.

C2: Valence electrons are localized standing-wave modes of the gauged Dirac operator on the vacuum torus.

C3: Interaction amplitudes are holonomies of the $U(1)$ connection along closed lattice cycles.

C4: Occupancy of a spatial channel is limited by the spinor suffix 10_2.

Conditional Theorem 11.2 (Two-mode bond: general spectral splitting)**.** Let ψ_A, ψ_B be two localized valent modes, taken orthonormal for simplicity. Restrict the gauged Dirac operator to their span:

$$D_A\big|_{\mathrm{span}\{\psi_A,\psi_B\}} = \begin{pmatrix} E_A & t_{AB} \\ t_{BA} & E_B \end{pmatrix}, \qquad t_{BA} = \overline{t_{AB}},$$

where

$$t_{AB} = \langle \psi_A | D_A | \psi_B \rangle = |t_{AB}| e^{i\Phi_{AB}},$$

and

$$\Phi_{AB} = \oint_{\gamma_{AB}} A \cdot dx \pmod{2\pi}.$$

The eigenvalues are

$$E_\pm = \frac{E_A + E_B}{2} \pm \sqrt{\left(\frac{E_A - E_B}{2}\right)^2 + |t_{AB}|^2}.$$

In particular,

$$E_- \leq \min(E_A, E_B), \qquad E_+ \geq \max(E_A, E_B).$$

If $E_A = E_B = E_0$, this reduces to

$$E_\pm = E_0 \pm |t_{AB}|.$$

The lower mode is bonding, the upper mode antibonding. For a closed cycle of interacting modes, constructive interference requires total holonomy closure:

$$\prod_{\text{cycle}} e^{i\Phi_{AB}} = \exp\left(i \oint A \cdot dx\right) = 1,$$

equivalently

$$\oint A \cdot dx \in 2\pi\mathbb{Z}.$$

Proof. The characteristic polynomial of the 2×2 Hermitian matrix is

$$\det \begin{pmatrix} E_A - E & t_{AB} \\ \overline{t_{AB}} & E_B - E \end{pmatrix} = (E_A - E)(E_B - E) - |t_{AB}|^2.$$

Solving gives

$$E_\pm = \frac{E_A + E_B}{2} \pm \sqrt{\left(\frac{E_A - E_B}{2}\right)^2 + |t_{AB}|^2}.$$

The inequalities follow from the Rayleigh–Ritz variational principle: the smallest eigenvalue is the minimum of the quadratic form over all normalized vectors, hence it is no larger than the value on either basis vector ψ_A or ψ_B; similarly for the largest eigenvalue.

The phase condition follows because the off-diagonal amplitudes are holonomies of the $U(1)$ connection along lattice cycles. A standing wave around a closed cycle is single-valued only if the total phase is an integer multiple of 2π. □

Proposition 11.3 (Pauli exclusion as register overflow)**.** *The subshell capacity is*

$$d_l = (l \ll 2) \vee 10_2.$$

Explicitly,

$$d_0 = 2, \quad d_1 = 6, \quad d_2 = 10, \quad d_3 = 14.$$

The suffix 10_2 *supplies exactly two spinor states per spatial channel. A third fermion in the same spatial channel would require a value* 3 *in a one-bit spin register, which is hardware overflow.*

Proof. Since $l \ll 2$ has two trailing binary zeros, bitwise OR with 10_2 appends the immutable spinor doublet. Thus each spatial mode has capacity two. No third orthogonal spinor bit exists in the register architecture. □

Numerology Guard 11.4. This is not a replacement for the spin-statistics theorem. It is the IT^3 architectural implementation of fermionic occupancy.

Proposition 11.5 (Octet as binary capacity)**.** *The active main-group valence register is $s \oplus p$. Its spinor-doubled capacity is*

$$d_0 + d_1 = 2 + 6 = 8.$$

Equivalently,

$$2(\dim s + \dim p) = 2(1 + 3) = 8.$$

Proof. Direct computation from $d_l = 2(2l + 1)$. □

Theorem 11.6 (Valence angles from O_h branching)**.** *The tetrahedral angle*

$$\theta = \arccos\left(-\frac{1}{3}\right) \approx 109.47^\circ$$

follows from orthogonality of four equivalent sp^3 hybrid channels inside

$$A_{1g} \oplus T_{1u}.$$

Proof. Let s be the normalized A_{1g} scalar mode and let p_x, p_y, p_z be an orthonormal basis of the T_{1u} vector mode. A normalized sp^3 hybrid pointing along a unit vector $u_i \in \mathbb{R}^3$ has the form

$$h_i = \frac{1}{2}s + \frac{\sqrt{3}}{2}(u_i \cdot p).$$

Its norm is

$$\|h_i\|^2 = \frac{1}{4} + \frac{3}{4} = 1.$$

Orthogonality of two distinct hybrids gives

$$0 = \langle h_i, h_j \rangle = \frac{1}{4} + \frac{3}{4}(u_i \cdot u_j).$$

Hence

$$u_i \cdot u_j = -\frac{1}{3}.$$

Therefore the angle between any two hybrid directions is

$$\theta = \arccos\left(-\frac{1}{3}\right).$$

□

hybridization	representation content	characteristic angle
sp	$A_{1g} \oplus T_{1u}$, one p-channel	180°
sp^2	$A_{1g} \oplus T_{1u}$, two p-channels	120°
sp^3	$A_{1g} \oplus T_{1u}$, three p-channels	109.47°
octahedral	d-channel participation	$90^\circ, 180^\circ$

Proposition 11.7 (Maximum pure p-covalency)**.** *Since*

$$\dim p = 3,$$

the maximum number of independent pure p-bonding channels is 3*. This gives the familiar single, double, triple bond hierarchy:*

$$\sigma, \quad \sigma + \pi, \quad \sigma + 2\pi.$$

Proof. Each bond channel consumes one spinor doublet. The p-block has exactly three orthogonal spatial channels. □

11.1 Block chemistry from binary capacities

Proposition 11.8 (s-block)**.** *The s-block capacity is*

$$d_0 = 2 = 10_2.$$

Chemically this is the scalar bit economy: donation or acceptance of one spinor doublet.

Proposition 11.9 (p-block)**.** *The p-block capacity is*

$$d_1 = 6 = 110_2.$$

Together with s, it gives the octet:

$$2 + 6 = 8.$$

Proposition 11.10 (d-block and the 18-electron rule)**.** *The d-block capacity is*

$$d_2 = 10 = 1010_2.$$

The full $s + p + d$ valence register is

$$2 + 6 + 10 = 18.$$

But

$$18 = 2 \cdot 3^2.$$

In binary,

$$3 = 11_2, \qquad 3^2 = 1001_2, \qquad 18 = 10010_2 = (3^2) \ll 1.$$

Thus the 18*-electron rule is the spinor-doubled square of the color bit* 3*.*

Proposition 11.11 (f-block and weak spatial directionality)**.** *The f-block corresponds to $\sqrt{5}$, orthogonal to the spatial torus T^3. Therefore pure f-transitions have*

$$\mathrm{ATFI} = 0.$$

This explains the weak spatial directionality and chemical similarity of lanthanide/actinide f-chemistry.

11.2 Krypton anomaly as binary carry

Proposition 11.12 (Krypton carry cascade)**.** *The closure sequence contains*

$$\mathrm{Ar} = 18, \qquad \mathrm{Kr} = 36.$$

Since

$$36 = 18 + 18,$$

and

$$18 = 0001\ 0010_2,$$

addition produces a binary carry:

$$0001\ 0010_2 + 0001\ 0010_2 = 0010\ 0100_2.$$

Equivalently,

$$\nu_2(18) = 1, \qquad \nu_2(36) = 2.$$

Thus the Krypton anomaly is a bitwise carry cascade, not a chemical miracle.

Proof. Direct binary arithmetic. Adding two equal numbers with 2-adic valuation 1 increases the valuation by at least one. □

12 The 16-bit vacuum processor: mask, instruction set, and binary traces

The number 16 is not postulated. It is the exact sum of the O_h spatial dimensions before spinor doubling:

$$\dim s + \dim p + \dim d + \dim f = 1 + 3 + 5 + 7 = 16.$$

The covolume theorem gives the static holographic mask

$$57\,600 = \texttt{0xE100} = \texttt{1110 0001 0000 0000}_2.$$

This section decodes that mask as a finite binary register and shows how the load-bearing arithmetic of the paper is executed inside it.

Numerology Guard 12.1 (Status of the processor language). The words ROM, ALU, instruction, register, and processor are architectural metaphors. The load-bearing facts are exact integer and binary identities. The physical claim that the vacuum executes these operations is Level B, conditional on the constructible-vacuum and topological-optics postulates.

12.1 The static 16-bit mask

Write the mask in nibbles:

$$\texttt{0xE100} = \texttt{1110 0001 0000 0000}_2.$$

nibble	bits	hex/binary	role
N_3	15–12	`0xE = 1110`	classical p-track plus equatorial barrier
N_2	11–8	`0x1 = 0001`	relativistic s-gateway
N_1	7–4	`0x0 = 0000`	dormant d-core
N_0	3–0	`0x0 = 0000`	dormant f-core

The active bits are exactly four:

$$\mathrm{wt}(\texttt{0xE100}) = 4.$$

They correspond to the active spatial valence dimensions

$$s + p = 1 + 3 = 4.$$

The dormant bits are twelve:

$$16 - 4 = 12,$$

corresponding to

$$d + f = 5 + 7 = 12.$$

After spinor doubling, the maximal floor capacity is

$$16 \cdot 2 = 32.$$

Useful masks are:

$$\begin{aligned} \texttt{ACTIVE_MASK} &= \texttt{0xE100}, \\ \texttt{P_TRACK_MASK} &= \texttt{0xE000}, \\ \texttt{BARRIER_BIT} &= \texttt{0x1000}, \\ \texttt{S_GATEWAY_BIT} &= \texttt{0x0100}, \\ \texttt{DORMANT_MASK} &= \texttt{0x1EFF}. \end{aligned}$$

Indeed,

$$\texttt{0xFFFF} \oplus \texttt{0xE100} = \texttt{0x1EFF},$$

and

$$\mathrm{wt}(\texttt{0x1EFF}) = 12.$$

12.2 Processor components

Component	Mathematical object	Binary realization
ROM	Covolume $\sqrt{\lvert\Delta_K\rvert} = 240^2$	`0xE100`
Opcode register	Squareclass group $\mathbb{F}_2^3$	3 bits `000`–`111`
Address bus	Floor n, block l, node μ	4+2+4 bits
Data bus	Spin σ, occupancy x	2 bits
ALU	XOR, left shift, bitwise OR, popcount	$\oplus$, $\ll$, $\vee$, wt
Flags	TFI, ATFI, LSB parity, halt	popcount, quadratic norm, mod 2
Clock	Pythagorean–Fibonacci cascade	$1, 2, 3, 5, 8, \ldots$
Halt	Kummer wall $\sqrt{13} \notin K$	illegal squareclass opcode

12.3 Squareclass instruction set

The radical opcode is a 3-bit instruction:

$$\mathrm{op}(d) = (b_5, b_3, b_2) \in \mathbb{F}_2^3, \qquad d = 2^{b_2} 3^{b_3} 5^{b_5}.$$

opcode	binary	radical d	meaning
`0x0`	`000`	1	compiled scalar unit
`0x1`	`001`	2	spinor bit $\sqrt{2}$
`0x2`	`010`	3	color bit $\sqrt{3}$
`0x3`	`011`	6	XOR: $2 \oplus 3$
`0x4`	`100`	5	curvature/f-bit $\sqrt{5}$
`0x5`	`101`	10	XOR: $2 \oplus 5$
`0x6`	`110`	15	XOR: $3 \oplus 5$
`0x7`	`111`	30	full register saturation

Any squarefree part outside

$$\{1, 2, 3, 5, 6, 10, 15, 30\}$$

is an illegal opcode. The first cascade illegal opcode is

$$13 = \texttt{0x0D},$$

because

$$\sqrt{13} \notin K.$$

12.4 Micro-operation table

Mnemonic	Semantic operation	Formula
`LOAD_SF`	squarefree compiler	$n = a^2 d \mapsto d$
`OPCODE`	3-bit radical opcode	$d \mapsto (b_5, b_3, b_2)$
`XOR_RAD`	radical multiplication	$\mathrm{op}(d_1 d_2) = \mathrm{op}(d_1) \oplus \mathrm{op}(d_2)$
`COMPILE`	squaring a radical	$(\sqrt{d})^2 = d \mapsto \mathrm{op} = 000$
`CAPACITY`	floor capacity	$C(q) = 2q^2 = (q^2) \ll 1$
`SPINOR_OR`	subshell capacity	$d_l = (l \ll 2) \vee 10_2$
`ALPHA_TFI`	binary friction	$\mathrm{wt}(Z \oplus (Z - 2))$
`ATFI2`	toroidal metric friction	$(\Delta n_s)^2 + 2(\Delta n_p)^2 + 3(\Delta n_d)^2$
`FACING`	equatorial mirror	$Z \mapsto F(Z)$
`MIRROR_FLOOR`	floor involution	$n \mapsto 11 - n$
`HALT_IF_WALL`	Kummer cutoff	if $\mathrm{sf}(n) \nmid 30$, halt

12.5 Binary execution traces

The following traces are exact arithmetic. They are not simulations.

Fano week boot.

$$\begin{aligned}
\mathrm{op}(2) \oplus \mathrm{op}(3) &= \texttt{001} \oplus \texttt{010} = \texttt{011} \rightsquigarrow 6, \\
\mathrm{op}(2) \oplus \mathrm{op}(5) &= \texttt{001} \oplus \texttt{100} = \texttt{101} \rightsquigarrow 10, \\
\mathrm{op}(3) \oplus \mathrm{op}(5) &= \texttt{010} \oplus \texttt{100} = \texttt{110} \rightsquigarrow 15, \\
\mathrm{op}(2) \oplus \mathrm{op}(3) \oplus \mathrm{op}(5) &= \texttt{001} \oplus \texttt{010} \oplus \texttt{100} = \texttt{111} \rightsquigarrow 30.
\end{aligned}$$

Kummer wall halt.

$$13 = \texttt{0x0D} = 1101_2$$

is not a valid 3-bit squareclass opcode. Hence

$$\sqrt{13} \notin K.$$

Covolume compilation.

$$1 \cdot 2 \cdot 3 \cdot 5 \cdot 8 = 240 = \texttt{0xF0},$$

and

$$\texttt{0xF0}^2 = \texttt{0xE100} = 57\,600.$$

Thus

$$\mathrm{wt}(\texttt{0xE100}) = 4, \qquad \mathrm{wt}(\texttt{0x1EFF}) = 12.$$

Capacity as shifted binary squares.

$$\begin{aligned}
q = 1: &\quad q^2 = \texttt{0001}, \quad C = q^2 \ll 1 = \texttt{0010} = 2, \\
q = 2: &\quad q^2 = \texttt{0100}, \quad C = q^2 \ll 1 = \texttt{1000} = 8, \\
q = 3: &\quad q^2 = \texttt{1001}, \quad C = q^2 \ll 1 = \texttt{10010} = 18, \\
q = 4: &\quad q^2 = \texttt{10000}, \quad C = q^2 \ll 1 = \texttt{100000} = 32.
\end{aligned}$$

Spinor subshell OR.

$$\begin{aligned} l=0: &\quad (0 \ll 2) \vee 10_2 = \texttt{0010} = 2,\\ l=1: &\quad (1 \ll 2) \vee 10_2 = \texttt{0110} = 6,\\ l=2: &\quad (2 \ll 2) \vee 10_2 = \texttt{1010} = 10,\\ l=3: &\quad (3 \ll 2) \vee 10_2 = \texttt{1110} = 14. \end{aligned}$$

Equatorial diode.

$$\begin{aligned} 88 &= \texttt{0x58} = \texttt{01011000}_2,\\ 86 &= \texttt{0x56} = \texttt{01010110}_2,\\ 88 \oplus 86 &= \texttt{0x0E} = \texttt{00001110}_2,\\ \mathrm{TFI}(88 \to 86) &= \mathrm{wt}(\texttt{0x0E}) = 3. \end{aligned}$$

And

$$\begin{aligned} 87 &= \texttt{0x57} = \texttt{01010111}_2,\\ 85 &= \texttt{0x55} = \texttt{01010101}_2,\\ 87 \oplus 85 &= \texttt{0x02} = \texttt{00000010}_2,\\ \mathrm{TFI}(87 \to 85) &= \mathrm{wt}(\texttt{0x02}) = 1. \end{aligned}$$

FCC closure.

$$43 = \texttt{0x2B} = \texttt{00101011}_2,$$

$$43 \ll 2 = \texttt{0xAC} = \texttt{10101100}_2 = 172.$$

Also

$$86 = \texttt{0x56}, \qquad 86 \ll 1 = \texttt{0xAC} = 172.$$

Fine-structure register.

$$13 = \texttt{0x0D}, \qquad 32 = \texttt{0x20},$$

$$13^2 = 169 = \texttt{0xA9},$$

$$169 - 32 = 137 = \texttt{0x89}.$$

Also

$$240 = \texttt{0xF0}, \qquad 103 = \texttt{0x67},$$

$$240 - 103 = 137 = \texttt{0x89}.$$

And

$$136 = \texttt{0x88}, \qquad 136 \vee 1 = \texttt{0x89} = 137.$$

Hourglass compression.

$$64 = \texttt{0x40} = \texttt{01000000}_2,$$

$$21 = \texttt{0x15} = \texttt{00010101}_2,$$

$$64 + 21 = 85 = \texttt{0x55} = \texttt{01010101}_2.$$

12.6 Hardware verification trace

The following trace is a deterministic audit of the finite register arithmetic used in the paper. Every line is either a Level A integer/binary identity or an explicitly flagged Level B architectural interpretation.

Step	Mnemonic	Binary/hex witness	Status
00	`INIT`	$R = \texttt{0x0000}$, op $= \texttt{000}$	A
01	`LOAD_UNIT`	$\mathrm{op}(1) = \texttt{000}$	A
02	`LOAD_SPINOR`	$\mathrm{op}(2) = \texttt{001}$, $\sqrt{2} \in K$	A
03	`LOAD_COLOR`	$\mathrm{op}(3) = \texttt{010}$, $\sqrt{3} \in K$	A
04	`LOAD_CURVATURE`	$\mathrm{op}(5) = \texttt{100}$, $\sqrt{5} \in K$	A
05	`XOR_FANO`	$\texttt{001} \oplus \texttt{010} = \texttt{011}$, $\texttt{001} \oplus \texttt{100} = \texttt{101}$, $\texttt{010} \oplus \texttt{100} = \texttt{110}$, $\texttt{001} \oplus \texttt{010} \oplus \texttt{100} = \texttt{111}$	A
06	`CASCADE`	$x = (1, 2, 3, 5, 8, 13, \ldots)$	A
07	`WALL_TEST`	$\mathrm{sf}(13) = 13 = \texttt{0x0D} = \texttt{0000 1101}_2$	A
08	`HALT_IF_WALL`	$13 \nmid 30 \Rightarrow \sqrt{13} \notin K$, illegal 3-bit opcode	A/B
09	`PRODUCT`	$1 \cdot 2 \cdot 3 \cdot 5 \cdot 8 = 240 = \texttt{0xF0}$	A
10	`SQUARE_COMPILE`	$\texttt{0xF0}^2 = \texttt{0xE100} = 57\,600$	A
11	`MASK_AUDIT`	$\mathrm{wt}(\texttt{0xE100}) = 4$, $\mathrm{wt}(\texttt{0x1EFF}) = 12$	A
12	`CAPACITY`	$\texttt{01} \ll 1 = \texttt{02}$, $\texttt{04} \ll 1 = \texttt{08}$, $\texttt{09} \ll 1 = \texttt{12}$, $\texttt{10} \ll 1 = \texttt{20}$	A
13	`SPINOR_OR`	$(0 \ll 2) \vee 2 = \texttt{02}$, $(1 \ll 2) \vee 2 = \texttt{06}$, $(2 \ll 2) \vee 2 = \texttt{0A}$, $(3 \ll 2) \vee 2 = \texttt{0E}$	A
14	`DIODE`	$\texttt{0x58} \oplus \texttt{0x56} = \texttt{0x0E}$, $\mathrm{wt} = 3$; $\texttt{0x57} \oplus \texttt{0x55} = \texttt{0x02}$, $\mathrm{wt} = 1$	A
15	`FACING`	$W_N(86) = \texttt{0x5676}$, $W_N(118) = \texttt{0x7656}$	A/B
16	`PHOTON_LATTICE`	$\|\vec{k}\|^2/(2\pi)^2 = n_1^2 + n_2^2/2 + n_3^2/3$	A under T^3 model
17	`REFLECT_CP`	attempted $\sqrt{13}$-tension $\mapsto$ mirror mode	B
18	`MASS_REGISTER`	incident + reflected standing wave $\mapsto$ rest mass	B
19	`BOND_XOR`	masked spinor-pair XOR $= \texttt{0x0020}$	B
20	`BOND_HOLONOMY`	$\prod e^{i\Phi_{AB}} = 1$	B

Proposition 12.2 (The Kummer wall is a register overflow)**.** *The squareclass register of* $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ *has exactly* $2^3 = 8$ *legal opcodes:*

$$\texttt{000}, \texttt{001}, \texttt{010}, \texttt{011}, \texttt{100}, \texttt{101}, \texttt{110}, \texttt{111}.$$

The cascade term $13 = \texttt{0x0D}$ *has squarefree part* 13*, which requires a prime bit outside* $\{2, 3, 5\}$*. Hence the wall*

$$\sqrt{13} \notin K$$

is exactly the statement that the 3-bit opcode register overflows.

Proof. By Theorem 2.2, legal squarefree parts are precisely

$$\{1, 2, 3, 5, 6, 10, 15, 30\},$$

which are encoded by the eight vectors of $\mathbb{F}_2^3$. Since 13 is prime and $13 \nmid 30$, it is not among them. In binary, $13 = \texttt{1101}_2$, so it cannot be represented by a 3-bit squareclass opcode. The cascade therefore halts at the first illegal allocation. □

12.7 Output artifacts of the register audit

Artifact	Exact output	Status
Fano week opcodes	`001`, `010`, `011`, `100`, `101`, `110`, `111`	A
Kummer halt	$13 =$ `0x0D` illegal in 3-bit squareclass register	A
Covolume mask	`0xE100` $= 57\,600 = 240^2$	A
Active/dormant split	4 active bits, 12 dormant bits	A
Capacity law	`02`, `08`, `12`, `20`	A
Subshell suffix	`02`, `06`, `0A`, `0E`	A
Equatorial diode	$\mathrm{TFI}(88 \to 86) = 3$, $\mathrm{TFI}(87 \to 85) = 1$	A
Facing byte-swap	`0x5676` $\leftrightarrow$ `0x7656`	A/B
Ogg boundary	$43 = \min(\mathcal{H} \setminus \mathcal{S}_M)$	A
FCC closure	$172 = 43 \ll 2$	A
Central edge	$86\|87 =$ `0x56`$\|$`0x57` $= 0\|1$, XOR $= 1$, sum $=$ $173 =$ `0xAD`	A
Low-nibble wall	`0xC` $= 12$ legal, `0xD` $= 13$ illegal	A
Central FCC cell	$85, 86, 87, 88$ pattern $1\,0\,1\,0$, checksum $346 =$ `0x15A`	A
Lichtenberg center	$43 = 2L_6+1$, $85 = L_8 = 4L_6+1$, $2^8 = 3L_8+1$	A
Central diode	$\mathrm{TFI}(88 \to 86) = 3$, $\mathrm{TFI}(87 \to 85) = 1$	A
Carry-free source	`0x56` + `0x57` = `0xAD` with no carry	A
Facing LSB no-go	$F(Z) \equiv Z \pmod 2$; one LSB bit is not CP	A
Spinor pair XOR	masked XOR = `0x0020`	B
Bond holonomy	$\prod e^{i\Phi_{AB}} = 1$	B

Thus the "simulation output" is not a fitted table of numbers. It is a finite set of exact register states, each either proved arithmetically or explicitly conditioned on the topological-optics postulates.

12.8 Binary constants of the framework

object	decimal	hex	binary
scalar unit	1	`0x01`	`0000 0001`
spinor seed	2	`0x02`	`0000 0010`
color bit	3	`0x03`	`0000 0011`
curvature bit	5	`0x05`	`0000 0101`
first plateau	8	`0x08`	`0000 1000`
Kummer wall	13	`0x0D`	`0000 1101`
equatorial capacity	32	`0x20`	`0010 0000`
classical hemisphere	86	`0x56`	`0101 0110`
absolute closure	172	`0xAC`	`1010 1100`
FCC cell count	43	`0x2B`	`0010 1011`
E_8 kissing number	240	`0xF0`	`1111 0000`
vacuum mask	57 600	`0xE100`	`1110 0001 0000 0000`
vacuum knot	103	`0x67`	`0110 0111`
fine-structure register	137	`0x89`	`1000 1001`
Heegner step	24	`0x18`	`0001 1000`
Lichtenberg L_6	21	`0x15`	`0001 0101`
Lichtenberg L_8	85	`0x55`	`0101 0101`
central antiphoton domain	86	`0x56`	`0101 0110`
central photon domain	87	`0x57`	`0101 0111`
source successor	173	`0xAD`	`1010 1101`
central cell checksum	346	`0x15A`	`1 0101 1010`

12.9 Noble-gas closures as binary register states

Z	hex	8-bit binary	structural event
2	`0x02`	`0000 0010`	He spinor seed
10	`0x0A`	`0000 1010`	Ne closure
18	`0x12`	`0001 0010`	Ar closure
36	`0x24`	`0010 0100`	Kr binary carry
54	`0x36`	`0011 0110`	Xe closure
86	`0x56`	`0101 0110`	Rn equator
118	`0x76`	`0111 0110`	Og closure
136	`0x88`	`1000 1000`	symmetry break
154	`0x9A`	`1001 1010`	relativistic filling
162	`0xA2`	`1010 0010`	penultimate approach
170	`0xAA`	`1010 1010`	alternating register
172	`0xAC`	`1010 1100`	absolute FCC closure

The final closure is a pure bit shift:

$$172 = \texttt{0xAC} = \texttt{0x56} \ll 1 = \texttt{0x2B} \ll 2.$$

12.10 How the 16-bit process works

The process is a strict pipeline.

1. **Boot from the scalar unit.** The empty register is `000`. The radicals $2, 3, 5$ load the three basis bits.

2. **Generate the Fano week.** XOR of basis bits produces the seven nonzero squareclass states:

$$2, 3, 6, 5, 10, 15, 30.$$

3. **Halt at the Kummer wall.** The cascade attempts $13 = \texttt{0x0D}$. This is not a valid 3-bit squareclass opcode. The processor halts:

$$\sqrt{13} \notin K.$$

4. **Compile the covolume.** The allowed cascade product is

$$1 \cdot 2 \cdot 3 \cdot 5 \cdot 8 = 240 = \texttt{0xF0}.$$

   Squaring compiles the radical register into the static mask:

$$\texttt{0xF0}^2 = \texttt{0xE100} = 57\,600.$$

5. **Load the 16-bit spatial ROM.** The mask declares four active spatial bits and twelve dormant bits:

$$\texttt{0xE100}: \quad 4 \text{ active}, \quad 12 \text{ dormant}.$$

6. **Fill floors by shifted squares.** The capacity law is a binary square followed by a left shift:

$$C(q) = (q^2) \ll 1.$$

7. **Route electrons by spinor OR.** Each spatial channel receives the immutable suffix $\texttt{10}_2$:

$$d_l = (l \ll 2) \vee 10_2.$$

8. **Evaluate nuclear friction.** Alpha-type transitions compute a borrow cascade:

$$\mathrm{TFI}(Z \to Z-2) = \mathrm{wt}(Z \oplus (Z-2)).$$

9. **Mirror across the equator.** The floor involution is

$$i(n) = 11 - n.$$

   The atomic facing map $F(Z)$ is a byte-swap partner in the nuclear facing word.

10. **Translate light into mass.** Under the topological-optics postulate, a photon mode reaching the Kummer wall reflects as its CP-conjugate mirror mode. The incident and reflected waves form a standing wave. Localized standing-wave energy is rest mass.

11. **Form chemical bonds.** Two occupied electronic words with the same spatial address and opposite spinor bit form a shared interference mode. Bonding requires holonomy closure:

$$\oint A \cdot dx \in 2\pi\mathbb{Z}.$$

13 Binary chemistry: electron, proton, atom, and molecule as register words

This section gives an explicit 16-bit register encoding for the operational roles of chemical objects. The encodings are architectural, not claims of known physical hardware. They are Level B unless the arithmetic itself is cited.

13.1 Electronic 16-bit word

Define an electronic register word by

$$W_e = (n \ll 12) \vee (l \ll 10) \vee (\mu \ll 6) \vee (\sigma \ll 5) \vee (x \ll 4) \vee (u \ll 1) \vee g,$$

where:

field	bits	meaning
n	15–12	topological floor, $0 \le n \le 11$
l	11–10	block, $0 = s, 1 = p, 2 = d, 3 = f$
μ	9–6	node index inside the block
σ	5	spinor bit
x	4	occupancy bit
u	3–1	auxiliary squareclass opcode tag
g	0	parity guard bit

The guard bit is

$$g = \mathrm{wt}(W_e^{\mathrm{raw}}) \bmod 2,$$

where W_e^{raw} is the word before appending g.

Proposition 13.1 (Spinor pair as a one-bit XOR)**.** *Let two electronic words have the same n, l, μ, u and occupancy $x = 1$, but opposite spin bits $\sigma = 0$ and $\sigma = 1$. Then, ignoring the guard bit,*

$$(W_e^{(0)} \oplus W_e^{(1)}) \ \textit{AND}\ \texttt{0xFFFE} = \texttt{0x0020}.$$

Thus a filled spatial channel is detected by one isolated spinor bit. A third fermion would require a second spin bit in the same address, which the register does not contain.

Proof. All fields except σ and possibly g are identical. The bit σ occupies position 5, i.e. mask `0x0020`. The guard bit can toggle because it is a parity of the upper bits, but it is masked out by `0xFFFE`. Hence the non-guard XOR is exactly `0x0020`. □

Numerology Guard 13.2. This is an architectural encoding of the spinor suffix 10_2. It is not a derivation of the spin-statistics theorem. It shows how the Pauli pair constraint appears as a one-bit register limitation.

13.2 Redox and alpha transitions as LSB operations

For an integer occupation register Z:

- one-electron transfer changes Z by ± 1 and toggles the LSB;
- two-electron transfer changes Z by ± 2 and preserves the LSB;
- alpha decay is a $\Delta Z = -2$ operation and preserves LSB parity.

Thus:

$$Z \mapsto Z \pm 1$$

is a redox-like parity switch, while

$$Z \mapsto Z - 2$$

is an alpha-like parity-preserving rail.

The binary friction of the alpha rail is

$$\mathrm{TFI}(Z \to Z - 2) = \mathrm{wt}(Z \oplus (Z - 2)).$$

13.3 Nuclear facing word

Let

$$\kappa(n) = \min(n, 11 - n), \qquad C(n) = 2 \left\lfloor \frac{\kappa(n) + 3}{2} \right\rfloor^2, \qquad \Sigma_n = \sum_{k=0}^{n} C(k).$$

This definition makes the capacity law explicitly palindromic:

$$C(11 - n) = C(n).$$

The cumulative closures are

$$(\Sigma_0, \ldots, \Sigma_{11}) = (2, 10, 18, 36, 54, 86, 118, 136, 154, 162, 170, 172).$$

For Z on floor n, meaning

$$\Sigma_{n-1} < Z \leq \Sigma_n$$

with $\Sigma_{-1} := 0$, define the facing map

$$F(Z) = Z + 172 - \Sigma_n - \Sigma_{n-1}.$$

Theorem 13.3 (Facing map is a fixed-point-free involution)**.** *The map F is a bijection of $\{1, \ldots, 172\}$, satisfies*

$$F(F(Z)) = Z,$$

has no fixed point, and preserves the intra-floor node index. In the symmetric IT^3 filling convention it preserves the block index. In particular F maps the classical half $\{1, \ldots, 86\}$ onto the relativistic half $\{87, \ldots, 172\}$.

Proof. The capacity law is palindromic by definition:

$$C(11-k) = C(k).$$

Hence

$$\Sigma_{11-n} = 172 - \Sigma_{n-1}, \qquad \Sigma_{10-n} = 172 - \Sigma_n.$$

If $\Sigma_{n-1} < Z \le \Sigma_n$, then

$$172 - \Sigma_n < F(Z) \le 172 - \Sigma_{n-1},$$

i.e.

$$\Sigma_{10-n} < F(Z) \le \Sigma_{11-n}.$$

Thus $F(Z)$ lies on floor $11-n$. Applying F again gives

$$F(F(Z)) = Z + 344 - (\Sigma_n + \Sigma_{n-1}) - (\Sigma_{11-n} + \Sigma_{10-n}) = Z.$$

Since F exchanges the two halves, $F(Z) \neq Z$. The node index is preserved because

$$F(Z) - \Sigma_{10-n} = Z - \Sigma_{n-1}.$$

The block index is preserved in the symmetric filling convention because the filling order on floor n and floor $11-n$ is identical by the capacity palindrome. □

Define the 16-bit nuclear facing word

$$W_N(Z) = (Z \ll 8) \vee F(Z).$$

Proposition 13.4 (Facing word is a byte-swap checksum)**.**

$$W_N(F(Z)) = \text{byteswap}(W_N(Z)).$$

Proof. By definition,

$$W_N(F(Z)) = (F(Z) \ll 8) \vee F(F(Z)) = (F(Z) \ll 8) \vee Z,$$

which is the byte swap of $(Z \ll 8) \vee F(Z)$. □

Z	$F(Z)$	$W_N(Z)$	interpretation
1	171	`0x01AB`	seed faces terminal relativistic node
2	172	`0x02AC`	He spinor seed faces absolute closure
56	88	`0x3858`	Ba faces Ra
86	118	`0x5676`	Rn faces Og
88	56	`0x5838`	Ra faces Ba
118	86	`0x7656`	Og faces Rn
172	2	`0xAC02`	absolute closure faces He seed

13.4 Central binary edge: photon as 1, antiphoton as 0, and the source at the Kummer wall

Let the absolute FCC closure be

$$N = 172 = 4 \cdot 43 = \texttt{0xAC},$$

and put

$$m = 43 = \texttt{0x2B}.$$

Define the two central nodes

$$A = m \ll 1 = 86 = \texttt{0x56}, \qquad P = (m \ll 1) \vee 1 = 87 = \texttt{0x57}.$$

If one uses the least significant bit as a crystal-domain label,

$$\pi(Z) = Z \bmod 2,$$

with the architectural reading

$$\pi = 1 \Rightarrow \text{photon domain}, \qquad \pi = 0 \Rightarrow \text{antiphoton domain},$$

then the centre of the binary crystal is not a point bit but the rising edge

$$A\,P = 0\,1.$$

The exact binary invariants of this central edge are

$$A \oplus P = \texttt{0x56} \oplus \texttt{0x57} = \texttt{0x01},$$

and

$$A + P = \texttt{0x56} + \texttt{0x57} = \texttt{0xAD} = 173 = N + 1.$$

The closure itself has low nibble

$$\texttt{0xAC}: \qquad \texttt{0xC} = 12,$$

and

$$\mathrm{sf}(12) = 3 \mid 30,$$

so 12 is a legal squareclass object. The central source successor has low nibble

$$\texttt{0xAD}: \qquad \texttt{0xD} = 13,$$

but

$$\mathrm{sf}(13) = 13 \nmid 30,$$

so 13 is the first illegal cascade opcode. Therefore

$$\texttt{0xAC} \longrightarrow \texttt{0xAD}$$

is exactly the one-bit move from the legal FCC closure to the Kummer wall.

Proposition 13.5 (Central edge exposes the Kummer wall)**.** *The unique central rising edge of the* 172*-node FCC register is*

$$86\,|\,87 = \texttt{0x56}\,|\,\texttt{0x57} = 0\,|\,1.$$

Its sum is

$$86 + 87 = 173 = \texttt{0xAD} = 172 + 1.$$

The low nibble of 172 *is* 12*, whose squarefree part is* $3 \mid 30$*, while the low nibble of* 173 *is* 13*, whose squarefree part is* $13 \nmid 30$*. Hence the centre of the crystal is the minimal binary successor of the absolute closure that reveals the algebraic wall* $\sqrt{13} \notin K$*.*

Proof. Direct hexadecimal arithmetic:

$$\texttt{0xAC} + 1 = \texttt{0xAD}.$$

The low nibbles are $\texttt{C} = 12$ and $\texttt{D} = 13$. Since

$$12 = 2^2 \cdot 3, \qquad \mathrm{sf}(12) = 3,$$

and $3 \mid 30$, the closure nibble is legal. Since 13 is prime and $13 \nmid 30$, the source-successor nibble is illegal. By Theorem 2.2, this is exactly the statement

$$\sqrt{13} \notin K.$$

□

object	hex	binary
central antiphoton domain A	`0x56`	`0101 0110`
central photon domain P	`0x57`	`0101 0111`
central edge XOR	`0x01`	`0000 0001`
source successor $A + P$	`0xAD`	`1010 1101`
closure low nibble	`0xC`	`1100`
wall low nibble	`0xD`	`1101`

Central FCC cell as a Lichtenberg cell

The closure 172 is exactly 43 FCC cells of four nodes:

$$172 = 43 \ll 2.$$

Since 43 is odd, there is a unique central cell. With one-based cell numbering, the central cell is

$$c = \frac{43 + 1}{2} = 22.$$

Its four nodes are

$$4c - 3,\ 4c - 2,\ 4c - 1,\ 4c = 85,\ 86,\ 87,\ 88.$$

In hexadecimal,

$$85 = \texttt{0x55}, \qquad 86 = \texttt{0x56}, \qquad 87 = \texttt{0x57}, \qquad 88 = \texttt{0x58}.$$

Their least-significant-bit pattern is

$$1\,0\,1\,0,$$

so the central FCC cell is a standing binary wave

$$P\,A\,P\,A.$$

The source is its middle edge

$$A\,P = 0\,1.$$

This cell is also the Lichtenberg cell. Recall

$$L_n = \left\lfloor \frac{2^n}{3} \right\rfloor, \qquad L_6 = 21, \qquad L_8 = 85.$$

Then

$$43 = 2L_6 + 1,$$

and

$$85 = L_8 = 4L_6 + 1.$$

Thus the zero-based central cell index is $L_6 = 21$, while the first node of the central cell is $L_8 = 85$. The central cell checksum is

$$85 + 86 + 87 + 88 = 346 = \texttt{0x15A}.$$

Equivalently,

$$\texttt{0x15A} = (\texttt{0x15} \ll 4) \vee \texttt{0x0A} = (L_6 \ll 4) \vee \texttt{0x0A},$$

and also

$$346 = 2 \cdot 173 = 2(N + 1).$$

Proposition 13.6 (The middle of the FCC crystal is a Lichtenberg cell)**.** *The central FCC cell of the* 172*-node closure is*

$$85, 86, 87, 88 = \texttt{0x55}, \texttt{0x56}, \texttt{0x57}, \texttt{0x58}.$$

Its LSB pattern is 1 0 1 0*, and its source edge is* 0 1*. Moreover,*

$$43 = 2L_6 + 1, \qquad 85 = L_8 = 4L_6 + 1,$$

so the central cell is indexed and initiated by Lichtenberg numbers.

Proof. Direct arithmetic:

$$L_6 = 21, \qquad 2L_6 + 1 = 43,$$

and

$$L_8 = 85, \qquad 4L_6 + 1 = 84 + 1 = 85.$$

The cell sum is

$$85 + 86 + 87 + 88 = 346 = 2 \cdot 173.$$

□

Central binary diode

The equatorial diode appears inside the central cell as a separation of the two LSB rails. For the even rail,

$$88 \to 86,$$

one has

$$88 = \texttt{0x58}, \qquad 86 = \texttt{0x56},$$

and therefore

$$88 \oplus 86 = \texttt{0x0E}, \qquad \mathrm{wt}(\texttt{0x0E}) = 3.$$

Thus

$$\mathrm{TFI}(88 \to 86) = 3.$$

For the odd rail,

$$87 \to 85,$$

one has

$$87 = \texttt{0x57}, \qquad 85 = \texttt{0x55},$$

and therefore

$$87 \oplus 85 = \texttt{0x02}, \qquad \mathrm{wt}(\texttt{0x02}) = 1.$$

Thus

$$\mathrm{TFI}(87 \to 85) = 1.$$

Hence the central cell is binary-birefringent:

$$\boxed{P\text{-rail } (\ldots 1) : \ \mathrm{TFI} = 1, \qquad A\text{-rail } (\ldots 0) : \ \mathrm{TFI} = 3.}$$

The odd photon-domain rail is transparent at the centre, while the even antiphoton-domain rail encounters a borrow cascade.

Proposition 13.7 (Central cell diode)**.** *Inside the central FCC cell,*

$$\mathrm{TFI}(88 \to 86) = 3, \qquad \mathrm{TFI}(87 \to 85) = 1.$$

Therefore the $P = 1$ *rail is smooth, while the* $A = 0$ *rail is refractive.*

Proof. Direct hexadecimal XOR and popcount:

$$\texttt{0x58} \oplus \texttt{0x56} = \texttt{0x0E}, \qquad \mathrm{wt}(\texttt{0x0E}) = 3,$$

and

$$\texttt{0x57} \oplus \texttt{0x55} = \texttt{0x02}, \qquad \mathrm{wt}(\texttt{0x02}) = 1.$$

□

The source is carry-free

The central edge satisfies

$$\texttt{0x56} + \texttt{0x57} = \texttt{0xAD}$$

without any hexadecimal carry:

$$6 + 7 = 13 = \texttt{D} < 16,$$

and

$$5 + 5 = 10 = \texttt{A} < 16.$$

Thus the source is a coherent carry-free superposition of the two central domains. This contrasts sharply with alpha-type friction, which is a borrow cascade:

$$\mathrm{TFI}(Z \to Z - 2) = \mathrm{wt}(Z \oplus (Z - 2)).$$

Proposition 13.8 (Carry-free central source)**.** *The central edge obeys*

$$\texttt{0x56} + \texttt{0x57} = \texttt{0xAD}$$

with no carry. Hence the binary source at the centre is a coherent sum, while nuclear friction is a borrow cascade.

Proof. Direct hexadecimal addition. □

The 16-bit mask already contains the central dipole

The vacuum mask is

$$\texttt{0xE100} = \texttt{1110 0001 0000 0000}_2.$$

At the byte boundary, bit 8 and bit 7 are

$$\mathrm{bit}_8 = 1, \qquad \mathrm{bit}_7 = 0.$$

Thus the isolated relativistic s-gateway bit is an island of 1 in a sea of 0. Its physical binary realization is not the point bit itself but its two edges:

$$0 \to 1, \qquad 1 \to 0.$$

Moreover,

$$2^8 = 256 = \texttt{0x0100},$$

and since

$$L_8 = 85,$$

one has the exact identity

$$2^8 = 3L_8 + 1.$$

Thus the gateway bit is tied to the colour bit 3, the Lichtenberg central node $L_8 = 85$, and the scalar seed 1.

Proposition 13.9 (Gateway-Lichtenberg identity)**.** *The relativistic gateway bit of the mask satisfies*

$$2^8 = 3L_8 + 1.$$

Equivalently,

$$\texttt{0x0100} = 3 \cdot \texttt{0x55} + 1.$$

Proof. Since $L_8 = \lfloor 256/3 \rfloor = 85$,

$$3L_8 + 1 = 255 + 1 = 256 = 2^8.$$

□

No-go: one LSB bit is not CP conjugation

The facing map preserves parity. Indeed, all capacities $C(n)$ are even, so all cumulative closures Σ_n are even. Therefore

$$F(Z) = Z + 172 - \Sigma_n - \Sigma_{n-1}$$

satisfies

$$F(Z) \equiv Z \pmod 2.$$

In particular,

$$F(86) = 118 \equiv 0 \pmod 2,$$

and

$$F(87) = 55 \equiv 1 \pmod 2.$$

Thus the geometric mirror does not turn the LSB label 1 into 0. If one uses

$$\pi(Z) = Z \bmod 2$$

as photon/antiphoton domain label, then the facing involution is a spatial mirror, not a charge-conjugation operator.

A genuine local CP flip requires a separate internal flag, say $\chi \in \{0, 1\}$, with

$$C : \chi \mapsto 1 - \chi,$$

while the geometric facing map acts as

$$F : \pi \mapsto \pi.$$

Proposition 13.10 (Facing preserves the one-bit domain label)**.** *For every $Z \in \{1, \ldots, 172\}$,*

$$F(Z) \equiv Z \pmod 2.$$

Hence the facing map does not swap the LSB photon/antiphoton domains. A one-bit LSB encoding is a crystal-domain order, not a CP-conjugation operator.

Proof. All $C(n)$ are even, hence all Σ_n are even. Since 172 is even,

$$F(Z) = Z + \text{even}$$

has the same parity as Z. □

Numerology Guard 13.11 (Photon/antiphoton encoding). In standard QED the photon is its own antiparticle. The present 1/0 assignment is not a claim about ordinary QED. It is an IT^3 architectural encoding of two crystal-domain states. The physical object is the edge 0 1 or 1 0, not an isolated bit. The global CP-like mirror is the free involution Γ on $\mathbb{S}^4 \times \mathbb{T}^3$, whose register shadow is the byte-swap of the nuclear facing word, not the LSB flip.

$$\begin{aligned}
\text{central antiphoton domain} &= 86 = \texttt{0x56},\\
\text{central photon domain} &= 87 = \texttt{0x57},\\
\text{central source edge} &= 0\,1,\\
\text{source sum} &= 173 = \texttt{0xAD} = 172 + 1,\\
\text{wall exposed in low nibble} &= \texttt{D} = 13,\\
\sqrt{13} \notin K &\Rightarrow \text{Kummer wall.}
\end{aligned}$$

13.5 Atom as a resonator word

An atom may be represented operationally as

$$\mathcal{A}_Z = W_N(Z) + \{W_e^{(1)}, \ldots, W_e^{(Z)}\},$$

where the electronic words are occupied according to the effective filling order. This is not a claim that the full many-body wavefunction is a 16-bit integer. It is a register-level abstraction:

- $W_N(Z)$ fixes the nuclear anchor and its equatorial mirror;
- the set of W_e words fills the valence register;
- chemical identity is determined by the last incomplete electronic register;
- molecular bonding is an interference relation between occupied electronic words.

Conditional Theorem 13.12 (Atom as optical resonator)**.** Conditional on the confined-light ontology, an atom is a topological optical resonator: a nuclear facing word plus a set of confined standing-wave electronic words. Its mass is localized wave energy; its chemistry is the spectrum of open valence modes.

13.6 Molecule as an interference graph

A molecule is a graph:

- vertices = atomic resonators $W_N(Z)$;
- edges = phase-closed shared electronic modes;
- edge labels = bond channels (n, l, μ);
- edge occupancy = spinor doublets;
- bond order = number of independent shared doublets.

Proposition 13.13 (Binary bond-order bound)**.** *A pure p-block bond order cannot exceed* 3 *unless s, d, or f channels participate. This is a direct consequence of the spatial dimension of the* T_{1u} *representation.*

Proof. Each bond channel consumes one spinor doublet. The p-block has exactly three orthogonal spatial channels. □

13.7 Summary of binary chemical roles

Entity	Binary object	Operational role
Electron	one bit σ inside W_e	mobile spinor carrier, pair completion, bond interference
Electron pair	XOR mask `0x0020` up to guard	filled spatial channel, Pauli saturation
Proton/nucleus	high byte of $W_N(Z)$	fixes atomic identity and register size
Facing partner	low byte of $W_N(Z)$	relativistic mirror across the equator
Atom	nuclear word plus occupied electronic words	optical resonator
Molecule	interference graph of electronic words	phase-closed bond network

Electron = mobile spinor bit,
Proton = fixed charge anchor,
Neutron = nuclear volume modifier,
Atom = optical resonator,
Molecule = interference graph,
Chemistry = phase-closed optics of confined light.

Numerology Guard 13.14. This section does not derive the proton mass, neutron stability, QCD confinement, or exact molecular energies. It gives the operational architecture: what each object is used for inside the topological-optics model. Numerical dynamics still require solving the gauged Dirac or spectral-action problem for the concrete register.

14 Topological closure of the facing involution on $\mathbb{S}^4 \times \mathbb{T}^3$

The spatial facing map

$$F(Z) = Z + 172 - \Sigma_n - \Sigma_{n-1}$$

was proved to be a fixed-point-free involution on $\{1, \ldots, 172\}$. Its topological origin is the free involution on the full vacuum arena

$$\mathcal{M}_{\text{tot}} = \mathbb{S}^4 \times \mathbb{T}^3.$$

Definition 14.1 (Free CP involution)**.** Let

$$a : \mathbb{S}^4 \to \mathbb{S}^4$$

be the antipodal map. Let

$$\tau : \mathbb{T}^3 \to \mathbb{T}^3$$

be a lattice involution chosen to implement the floor reflection

$$i(n) = 11 - n$$

on the finite floor-label model of the torus. Define

$$\Gamma : \mathcal{M}_{\text{tot}} \to \mathcal{M}_{\text{tot}}, \qquad \Gamma(s, \xi) = (a(s), \tau(\xi)).$$

Proposition 14.2. Γ *is a free involution.*

Proof. The antipodal map a on $\mathbb{S}^4$ has no fixed point. Therefore, if $\Gamma(s, \xi) = (s, \xi)$, then $a(s) = s$, impossible. Also $\Gamma^2 = (a^2, \tau^2) = (\text{id}, \text{id})$, because $a^2 = \text{id}$ and $i(i(n)) = n$. Hence Γ is a free involution. □

Proposition 14.3 (Descent of the 172-state bundle)**.** *Because* Γ *is free, the* 172*-state topological bundle descends to an* 86*-state bundle on the quotient*

$$\mathcal{M}_{\text{tot}}/\Gamma.$$

This is the geometric reason why the classical and relativistic hemispheres are paired by

$$86 + 86 = 172.$$

Proof. The floor involution $i(n) = 11 - n$ has no integer fixed point because $2n = 11$ has no solution in $\mathbb{Z}$. Thus the 12 floors split into six free orbits. The total number of states is doubled by spinor occupancy and hemisphere duplication, giving 172 states. The free involution Γ identifies opposite hemispherical states, leaving $172/2 = 86$ quotient states. □

Proposition 14.4 (Action on instanton and holonomy sectors)**.** *The involution* Γ *acts as CP conjugation on the two factors:*

- *on* $\mathbb{S}^4$*, the antipodal map reverses orientation; for an* $SU(2)$ *instanton sector with second Chern number* k*, it sends*
$$k \longmapsto -k;$$
- *on* $\mathbb{T}^3$*, the flat* $U(1)$ *holonomy class* $[v]$ *is sent to its inverse:*
$$[v] \longmapsto [v]^{-1}.$$

Therefore, for the external Kasparov product

$$u = \beta \boxtimes v \in K_1(\mathbb{S}^4 \times \mathbb{T}^3),$$

where $\beta \in \widetilde{K}^0(\mathbb{S}^4) \simeq \mathbb{Z}$ *is the Bott generator, one has, up to the standard sign convention,*

$$\Gamma^*[u] = -[u]^{-1}.$$

The absolute Toeplitz index is invariant:

$$|\operatorname{Ind}(P_u P)| = 86.$$

Sketch. The antipodal map on $\mathbb{S}^4$ has degree -1, hence acts by multiplication by -1 on the Bott generator of $\widetilde{K}^0(\mathbb{S}^4)$. Complex conjugation of a flat $U(1)$ connection inverts its holonomy class in $K_1(\mathbb{T}^3)$. Functoriality of the external Kasparov product gives the stated transformation. The APS/Toeplitz index changes sign under this conjugation, but its absolute value, which counts the classical hemisphere volume, remains 86. □

Remark 14.5 (Byte-swap as the register shadow of Γ). The nuclear facing word

$$W_N(Z) = (Z \ll 8) \vee F(Z)$$

satisfies

$$W_N(F(Z)) = \text{byteswap}(W_N(Z)).$$

This is the finite 16-bit shadow of the free involution Γ: the high byte and low byte are exchanged exactly as the classical and relativistic hemispheres are exchanged. The $\mathbb{S}^4$ factor supplies the topological sign flip that makes the involution free, while the $\mathbb{T}^3$ factor carries the discrete floor/node coordinates.

Numerology Guard 14.6. The freeness of Γ, the descent to 86 quotient states, and the byte-swap identity are Level A topology/arithmetic. The interpretation of Γ as CP conjugation of confined light is Level B, conditional on the topological-optics ontology. No dynamical derivation of instanton masses or QCD confinement is claimed.

15 The 137 register, FCC closure $Z = 172$, and the Ogg boundary

Proposition 15.1 (Exact identities)**.** *One has*

$$137 = 13^2 - 32,$$

$$137 = 240 - 103,$$

and in binary

$$136 = 1000\ 1000_2, \qquad 137 = 1000\ 1001_2 = 136 \vee 1.$$

The Hamming weight is

$$\text{wt}(137) = 3.$$

Proof. Direct arithmetic:

$$13^2 - 32 = 169 - 32 = 137,$$

$$240 - 103 = 137.$$

The binary decomposition is immediate. □

Numerology Guard 15.2. The identities are Level A. The identification

$$137 \approx \alpha^{-1}$$

is Level C resonance unless a separate dynamical derivation is supplied.

Theorem 15.3 (Binary closure shifts)**.** *The classical hemisphere sum is*

$$\sum_{n=0}^{5} C(n) = 86.$$

The full closure is

$$172 = 86 + 86.$$

In binary,

$$86 = 0101\ 0110_2, \qquad 172 = 1010\ 1100_2.$$

Hence

$$172 = 86 \ll 1.$$

Also

$$172 = 43 \cdot 4 = 43 \ll 2.$$

Proof. The capacity law gives

$$C(0), \ldots, C(5) = 2, 8, 8, 18, 18, 32,$$

whose sum is 86. Doubling gives 172. Binary shifting left by one multiplies by 2, and shifting 43 left by two multiplies by 4. □

15.1 Monster Moonshine and the Ogg boundary of the FCC closure

The number 43 in the closure

$$172 = 43 \ll 2$$

is not an arbitrary cell count. It is the first Heegner prime that lies outside the Monster-prime set. This is a theorem of Ogg's Moonshine classification, not a numerical coincidence.

Theorem 15.4 (Ogg–Heegner boundary)**.** *Let*

$$\mathcal{H} = \{2, 3, 7, 11, 19, 43, 67, 163\}$$

be the Heegner primes, i.e. the primes p for which $\mathbb{Q}(\sqrt{-p})$ has class number one. Let

$$\mathcal{S}_M = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 41, 47, 59, 71\}$$

be the set of prime divisors of the order of the Monster group M. Then

$$\mathcal{H} \cap \mathcal{S}_M = \{2, 3, 7, 11, 19\},$$

and therefore

$$\boxed{43 = \min(\mathcal{H} \setminus \mathcal{S}_M).}$$

Equivalently, 43 *is the first Heegner prime for which the Moonshine genus is nonzero.*

Proof. By Ogg's theorem, the primes dividing the Monster order are exactly the supersingular primes, equivalently those primes p for which the Atkin–Lehner quotient modular curve $X_0(p)^+$ has genus zero [7]. The Heegner primes are the classical Stark–Heegner list

$$2, 3, 7, 11, 19, 43, 67, 163.$$

Intersecting this list with the Monster-prime set gives

$$\{2, 3, 7, 11, 19\}.$$

The next Heegner prime is 43, and $43 \notin \mathcal{S}_M$. Hence 43 is the first Heegner prime outside the Monster-prime set. By Ogg's theorem again, $X_0(43)^+$ does not have genus zero; in fact its genus is one. □

Proposition 15.5 (The FCC closure hits the Ogg boundary exactly)**.** *The absolute vacuum closure is*

$$172 = 86 \ll 1 = 43 \ll 2.$$

Thus the macroscopic FCC crystal closes in exactly 43 *unit cells. Since* 43 *is the first Ogg-breaking Heegner prime, the closure of the periodic system occurs precisely at the first arithmetic obstruction to genus-zero Moonshine symmetry.*

Proof. Direct binary arithmetic:

$$43 = \texttt{0x2B}, \qquad 43 \ll 2 = \texttt{0xAC} = 172.$$

The theorem above identifies 43 as the first Heegner prime outside the Monster-prime set. □

Remark 15.6 (Singular modulus bridge). The Heegner module attached to 43 has exact singular modulus

$$j\left(\frac{1+\sqrt{-43}}{2}\right) = -960^3.$$

Moreover,

$$960 = 4 \cdot 240 = 2 \cdot 480.$$

Thus the same integer 4 that appears as the number of FCC nodes per cell also appears in the bridge between the Heegner cell count and the E_8 kissing number:

$$43 \text{ cells} \quad \longleftrightarrow \quad j(\sqrt{-43}) = -960^3, \quad 960 = 4 \cdot 240.$$

All identities in this remark are exact theorems of complex multiplication and sphere-packing arithmetic.

Remark 15.7 (Noncommutative-geometric reading). In Connes' noncommutative geometry, rigid arithmetic data can be encoded through finite-dimensional algebras, K-theory classes, and spectral triples [4]. The present construction does not derive the Monster group from the vacuum. Rather, it identifies an arithmetic boundary: the FCC closure of the IT^3 vacuum selects the first Heegner cell that is not absorbed by the genus-zero Moonshine set. This is the strongest honest statement: the number 43 is not a decorative coincidence; it is the exact Ogg boundary of the closure.

Numerology Guard 15.8. Level A: Ogg's theorem, the Heegner list, the Monster-prime list, $172 = 43 \ll 2$, and the singular modulus identity. Level C: the physical interpretation that the vacuum crystal "breaks Monster symmetry" at the 43rd cell. No group-theoretic derivation of nuclear stability from the Monster is claimed.

16 Ledger: theorem, conditional theorem, resonance, open target

Statement	Status
$\sqrt{n} \in K \iff \mathrm{sf}(n) \mid 30$	Level A
Seven quadratic subfields $= \mathbb{F}_2^3 \setminus \{0\}$	Level A
$\Delta_K = 2^{16}3^4 5^4$	Level A
$\sqrt{\lvert\Delta_K\rvert} = 57\,600 = 240^2$	Level A
$57\,600 = \texttt{0xE100} = 1110\ 0001\ 0000\ 0000_2$	Level A
$\mathrm{wt}(\texttt{0xE100}) = 4$, dormant weight $= 12$	Level A
Rank-three derivation of K from T^3, orthogonal composition, cascade, and maximal delay	Level A conditional
Uniqueness of $\langle 2, 3, 5\rangle$ as maximal rank-three squareclass group	Level A conditional
Optimal rank-three cascade delay: no rank-three register passes 13	Level A conditional
Cascade halts at 13	Level A
Wall selectivity: $\sqrt{15}, \sqrt{30} \in K$, but $\sqrt{13} \notin K$	Level A
$1 \cdot 2 \cdot 3 \cdot 5 \cdot 8 = 240$	Level A
Cohn: only nontrivial Fibonacci square is $144 = 12^2$	Level A (cited theorem)
If global scale is Fibonacci-square, then $L = 12$	Level B
Heegner primes $p > 11$ satisfy $p \equiv 19 \pmod{24}$	Level A
Exact 2-adic TFI formula	Level A
Binary diode for $E \equiv 6 \pmod 8$	Level A
Subtractive Pascal parity $S(n,k) \equiv \binom{n}{k} \pmod 2$	Level A
Hourglass row balance $\sum_k S(n,k) = 2$	Level A
Lichtenberg compression $L_{n+2} = 2^n + L_n$	Level A
ATFI as toroidal metric norm	Level A under metric postulate
Discrete photon wavevectors on T^3	Level A under compact-torus model
Additive cascade wall by $\sqrt{13} \notin K$	Level B under constructibility
Mass as confined light	Level B under ontology
Chemical bond as phase-closed interference	Level B under chemical optics
Pauli as register overflow	Level B architectural reinterpretation
Octet $= 2(1+3)$	Level A arithmetic, Level B mapping
Tetrahedral angle from sp^3 orthogonality	Level A geometry, Level B mapping
Facing map F is fixed-point-free involution	Level A
Nuclear facing word byte-swap	Level A arithmetic, Level B encoding
Electronic 16-bit word	Level B architecture
Spinor pair XOR `0x0020` up to guard	Level B architecture
16-bit processor ISA	Level B analogy
Execution trace `0xE100-BOOT`	Level A tool
Kummer wall as 3-bit register overflow	Level A
Covolume compilation $\texttt{0xF0}^2 = \texttt{0xE100}$	Level A
Ogg–Heegner boundary $43 = \min(\mathcal{H} \setminus \mathcal{S}_M)$	Level A
Singular modulus $j(\sqrt{-43}) = -960^3$, $960 = 4 \cdot 240$	Level A
FCC closure hits Ogg boundary	Level A arithmetic, Level C interpretation
Free involution Γ on $\mathbb{S}^4 \times \mathbb{T}^3$	Level A topology
Byte-swap as register shadow of Γ	Level A arithmetic, Level B encoding
CP reading of Γ	Level B under confined-light ontology
Central edge 86|87 = `0x56`|`0x57`, XOR $= 1$, sum $= 173 = \texttt{0xAD}$	Level A
Low-nibble audit `0xC` $= 12$ legal, `0xD` $= 13$ illegal	Level A
Central FCC cell $85, 86, 87, 88$ pattern $1\,0\,1\,0$, checksum $346 = \texttt{0x15A}$	Level A
Lichtenberg central cell $43 = 2L_6 + 1$, $85 = L_8 = 4L_6 + 1$	Level A
Central diode $\mathrm{TFI}(88 \to 86) = 3$, $\mathrm{TFI}(87 \to 85) = 1$	Level A
Carry-free source $\texttt{0x56} + \texttt{0x57} = \texttt{0xAD}$	Level A
Mask central dipole $\mathrm{bit}_8 = 1$, $\mathrm{bit}_7 = 0$ in `0xE100`	Level A

17 Conclusion

The strict core of the framework is the following chain of theorems:

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}) \longrightarrow \mathbb{F}_2^3 \longrightarrow \text{seven quadratic subfields} \longrightarrow \Delta_K = 2^{16}3^4 5^4 \longrightarrow \sqrt{|\Delta_K|} = 240^2 \longrightarrow 57\,600_{10} =$$

Parallelly,

$$1, 2, 3, 5, 8 \longrightarrow 240, \qquad 13 \longrightarrow \sqrt{13} \notin K \longrightarrow \text{additive cascade wall.}$$

The field K is no longer an arbitrary prime assignment: under the rank-three toroidal register, orthogonal composition, Pythagorean–Fibonacci cascade, and maximal-constructibility principle, it is forced as the unique rank-three squareclass field that delays the first new prime as long as possible. Moreover, Theorem 3.6 shows that no rank-three squareclass register can do better: the wall at 13 is optimal.

The binary friction laws, the Heegner step 24, the equatorial diode, the octet, the 18-electron rule, the tetrahedral angle, the facing byte-swap, the Krypton carry, the FCC shift $172 = 43 \ll 2$, the Ogg boundary, and the 137 register are then either exact arithmetic consequences or conditional theorems under explicitly stated optical/topological postulates.

The register trace converts the alleged "numerology" into a finite-state audit: the vacuum does not guess numbers; it allocates opcodes. The wall $\sqrt{13} \notin K$ is not a mystical cutoff but a 3-bit overflow. The mask `0xE100` is not a decorative binary string but the compiled covolume. The closure $172 = 43 \ll 2$ is not a lucky factorization but the first Ogg-breaking Heegner boundary. Chemistry then appears as the interference of the modes that this finite register allows.

The central-edge audit sharpens this further. The middle of the 172-node crystal is the carry-free dipole

$$\texttt{0x56} \,|\, \texttt{0x57} = 0 \,|\, 1,$$

whose sum is

$$\texttt{0xAD} = 172 + 1.$$

Its low nibble is 13, the first illegal squareclass opcode. Hence the source at the centre of the crystal is not an additional metaphysical postulate: it is the minimal binary successor of the absolute closure that exposes the Kummer wall. The photon/antiphoton 1/0 labelling is therefore consistent only as a domain-edge encoding; the facing involution preserves the least significant bit, while genuine CP conjugation requires the separate topological involution on $\mathbb{S}^4 \times \mathbb{T}^3$ or an independent internal flag.

The appendices formalize this optical-domain reading as an address/data architecture, provide a machine-checkable register audit, and establish the first numerical bridge from the toroidal spectrum to laboratory units:

$$T_{\min} = \frac{\mathrm{Ry}}{3} \approx 4.535 \text{ eV} \quad \text{vs.} \quad D_0(\mathrm{H}_2) \approx 4.478 \text{ eV}.$$

Appendix F records five explicit falsification bets that distinguish the framework from post-hoc numerology. Appendix G then converts the remaining gap between arithmetic architecture and physics into four explicit Level O research targets.

The final ontological formula, with its epistemic status preserved, is:

$$
\begin{aligned}
\text{Vacuum} &= O_h\text{-crystal over } K,\\
\text{Root} &= \text{opcode},\\
\text{Square} &= \text{compile},\\
\text{XOR} &= \text{radical multiplication},\\
\sqrt{13} \notin K &= \text{optimal cascade wall},\\
\text{Mask} &= \texttt{0xE100},\\
\text{Central edge} &= \texttt{0x56}|\texttt{0x57} = 0|1,\\
\text{Source sum} &= \texttt{0xAD} = 172 + 1,\\
\text{Hourglass} &= \text{fractal parity register},\\
\text{Light} &= \text{discrete torus harmonic},\\
\text{Mass} &\overset{\text{postulate}}{=} \text{confined light},\\
\text{Atom} &\overset{\text{postulate}}{=} \text{optical resonator},\\
\text{Bond} &\overset{\text{postulate}}{=} \text{phase-closed interference},\\
\text{Chemistry} &\overset{\text{postulate}}{=} O_h\text{-optics of confined light.}
\end{aligned}
$$

In one sentence:

Chemistry is the interference pattern of light that the algebraic wall $\sqrt{13} \notin K$ refused to let escape.

A Address/Data duality of proton and electron

This appendix provides the strict architectural translation of atomic components into the 16-bit topological register. It eliminates the continuous "particle in a box" metaphor, replacing it with a deterministic Address/Data bus architecture.

Theorem A.1 (Hardware/Software Duality of Nucleus and Electron)**.** *In the IT^3 binary framework, the proton count Z and the electron occupancy operate on fundamentally different computational axes:*

- ***Proton (Z) is the Hardware Address Pointer:*** *It dynamically allocates the topological nodes by setting the boundary $\Sigma_{n-1} < Z \leq \Sigma_n$.*
- ***Electron (e^-) is the Software Data Payload:*** *It is a Boolean state vector (occupancy $x \in \{0, 1\}$, spin $\sigma \in \{0, 1\}$) executed at the allocated addresses.*

Proof. By the palindromic capacity law

$$C(n) = 2 \left\lfloor \frac{\min(n, 11 - n) + 3}{2} \right\rfloor^2,$$

the lattice geometry is fixed independently of the particles. The integer Z determines exactly how many nodes are topologically active, i.e. the size of the instantiated register array. An electron does not alter the geometric capacity; it toggles the x (occupancy) and σ (spinor) bits of the electronic word W_e. Therefore, altering Z changes the physical hardware geometry (transmutation), while altering e^- changes only the logic state of the existing geometry (ionization). □

Entity	Register role	Binary witness
Proton / nucleus	Address pointer and register allocator	high byte of $W_N(Z) = (Z \ll 8) \vee F(Z)$
Facing partner	Mirror address across equator	low byte of $W_N(Z)$
Electron	Data bit at fixed address	σ, x inside W_e
Electron pair	Filled data channel	masked XOR = `0x0020`

Numerology Guard A.2. This is an architectural Level B statement. It does not derive QCD, nuclear binding energies, or the proton mass. It separates two operations that are often conflated: changing the nuclear address Z versus changing the electronic data state.

B Topological optics: the π-phase shift and LSB inversion

We previously established the central binary edge

$$86 \,|\, 87 = \texttt{0x56} \,|\, \texttt{0x57} = 0 \,|\, 1$$

as the transition between the antiphoton and photon domains. We now prove that, under the confined-light reflection postulate, this edge is the discrete shadow of an optical phase reversal.

Conditional Theorem B.1 (Optical reflection as binary inversion)**.** Assume the Confined-Light Ontology: matter is light undergoing total internal reflection at the Kummer wall $\sqrt{13} \notin K$. In classical and quantum optics, reflection at an absolutely dense boundary induces a phase shift of π. The phase multiplier is

$$e^{i\pi} = -1.$$

Introduce a binary phase flag $\varphi \in \mathbb{F}_2$ by

$$\text{sign} = (-1)^{\varphi}.$$

Reflection multiplies the sign by -1, hence

$$\varphi \longmapsto \varphi \oplus 1.$$

Proposition B.2 (The LSB flip is the discrete shadow of the optical π-shift)**.** *If the local crystal-domain label is identified with the phase flag,*

$$\varphi = \pi(Z) = Z \bmod 2,$$

then optical reflection acts as

$$\pi(Z) \longmapsto \pi(Z) \oplus 1.$$

For the central edge,

$$\pi(87) = 1, \qquad \pi(86) = 0,$$

and

$$87 \oplus 1 = 86, \qquad 86 \oplus 1 = 87.$$

Thus the local domain inversion $1 \leftrightarrow 0$ *is the exact binary execution of a* π*-phase reversal.*

Proof. Modulo 2,

$$-1 \equiv 1 \pmod 2.$$

Therefore multiplication of the sign by -1 is addition of 1 in $\mathbb{F}_2$. The LSB map

$$\pi(Z) = Z \bmod 2$$

is a homomorphism from integer parity to $\mathbb{F}_2$. Hence a sign flip is represented by toggling the LSB. Since 86 and 87 differ exactly by one in the LSB and

$$86 \oplus 87 = \texttt{0x01},$$

the central edge is the minimal binary realization of the phase reversal. □

Numerology Guard B.3 (Critical no-go). This proposition does *not* say that the facing map F is the CP conjugation. In fact,

$$F(Z) \equiv Z \pmod 2,$$

so the geometric mirror preserves the LSB. The π-shift statement is a local optical-domain identification, not a global charge-conjugation operator. A genuine CP flip requires either the free involution Γ on $\mathbb{S}^4 \times \mathbb{T}^3$ or a separate internal flag χ.

$$\begin{aligned}
\text{optical reflection} &\Rightarrow \varphi \mapsto \varphi \oplus 1,\\
\text{local domain label} &\Rightarrow \pi = Z \bmod 2,\\
\text{central edge} &\Rightarrow 87 \leftrightarrow 86,\\
\text{binary witness} &\Rightarrow 87 \oplus 1 = 86,\\
\text{facing map} &\Rightarrow F(87) = 55 \equiv 1, \quad F(86) = 118 \equiv 0.
\end{aligned}$$

C Compact register trace sheet

The following trace sheet collects the load-bearing hexadecimal identities used in the main text and appendices. It is not a simulation. Every line is an exact integer/binary identity, except where explicitly marked as an architectural reading.

```
DERIVATION OF K
  rank Gamma = dim H^1(T^3;F2)=3
  cascade sf: 1,2,3,5,2,13
  to pass 1,2,3,5 need independent bits 2,3,5
  rank 3 => Gamma=<2,3,5>
  first overflow = 13

WALL SEMANTICS
  15 in {1,2,3,5,6,10,15,30} => sqrt(15) in K
  30 in {1,2,3,5,6,10,15,30} => sqrt(30) in K
  13 not in allowed squareclasses => sqrt(13) not in K
  wall is additive cascade overflow, not squareclass ceiling

BOOT FANO WEEK
  op(1)=000, op(2)=001, op(3)=010, op(5)=100
  XOR:
    001 ^ 010 = 011 -> 6
    001 ^ 100 = 101 -> 10
    010 ^ 100 = 110 -> 15
    001 ^ 010 ^ 100 = 111 -> 30

WALL
  sf(13)=13=0x0D
  0x0D not in {0x0,0x1,0x2,0x3,0x4,0x5,0x6,0x7}
  HALT: sqrt(13) not in K

COMPILE COVOLUME
  allowed cascade product = 1*2*3*5*8 = 240 = 0xF0
  mask = 0xF0^2 = 0xE100 = 57600
  popcount(0xE100)=4
  popcount(0x1EFF)=12
```

```
PALINDROMIC CAPACITY
  kappa(n)=min(n,11-n)
  C(n)=2*floor((kappa(n)+3)/2)^2
  C = [2,8,8,18,18,32,32,18,18,8,8,2]
  sum(first 6)=86, sum(all)=172

EQUATORIAL DIODE
  88 = 0x58, 86 = 0x56
  0x58 ^ 0x56 = 0x0E, popcount = 3

  87 = 0x57, 85 = 0x55
  0x57 ^ 0x55 = 0x02, popcount = 1

FCC CLOSURE
  43 = 0x2B
  43 << 2 = 0xAC = 172
  86 = 0x56
  86 << 1 = 0xAC = 172

FINE-STRUCTURE REGISTER
  13 = 0x0D, 32 = 0x20
  13^2 = 169 = 0xA9
  169 - 32 = 137 = 0x89
  240 = 0xF0, 103 = 0x67
  240 - 103 = 137 = 0x89
  136 = 0x88, 136 | 1 = 0x89 = 137

CENTRAL BINARY EDGE
  A = 86 = 0x56 = 01010110
  P = 87 = 0x57 = 01010111
  A XOR P = 0x01
  A + P = 0xAD = 173 = 172 + 1
  low nibble 0xAC = 0xC = 12, sf(12)=3 legal
  low nibble 0xAD = 0xD = 13, sf(13)=13 illegal

CENTRAL FCC CELL
  85 = 0x55, 86 = 0x56, 87 = 0x57, 88 = 0x58
  LSB pattern = 1 0 1 0
  source edge = 0 1
  checksum = 85+86+87+88 = 346 = 0x15A = 2*173

LICHTENBERG CENTER
  L6 = 21 = 0x15
  L8 = 85 = 0x55
  43 = 2*L6 + 1
  85 = 4*L6 + 1
  2^8 = 3*L8 + 1 = 256 = 0x0100

H2 BRIDGE
  D^2 spectrum on T^3: n1^2 + n2^2/2 + n3^2/3
  lowest non-scalar denominator = 3
  Ry/3 = 13.605693/3 = 4.535231 eV
```

```
  D0(H2) approx 4.478 eV
  relative deviation approx 1.3%

OPTICAL PHASE BIT
  pi(86)=0, pi(87)=1
  phase flip: pi -> pi XOR 1
  87 -> 86 is local domain inversion
  facing F preserves pi:
    F(86)=118 (0), F(87)=55 (1)

ADDRESS/DATA
  W_N(86)=0x5676, W_N(118)=0x7656 = byteswap
  spinor pair XOR mask = 0x0020
```

D Machine-checkable binary auditor

The following Python script audits the 16-bit mask, the squareclass compiler, the rank-three derivation and optimality of K, the wall semantics, the palindromic capacity law, the facing map, the binary diode, the Heegner step, the Ogg boundary, the Perez Hourglass parity core, the Lichtenberg compression identity, the Central Edge `0xAD`, the optical phase-bit witness, the first H_2 numerical bridge, and the proposed register words.

Numerology Guard D.1 (Status of the auditor). The script is a Level A verification tool for exact integer and binary identities. It does not simulate physical dynamics. Level B architectural interpretations are not asserted by the script; they are only accompanied by the arithmetic witnesses they use. Level O targets are not audited as theorems.

```
import math
from itertools import combinations

def popcount(x):
    return bin(x).count("1")

def nu2(m):
    assert m > 0
    t = 0
    while m % 2 == 0:
        m //= 2
        t += 1
    return t

def squarefree_part(n):
    d = 1
    p = 2
    m = n
    while p * p <= m:
        count = 0
        while m % p == 0:
            m //= p
            count += 1
        if count % 2 == 1:
            d *= p
        p += 1 if p == 2 else 2
```

```
    if m > 1:
        d *= m
    return d

def byteswap16(x):
    return ((x & 0x00FF) << 8) | ((x & 0xFF00) >> 8)

def span_set(primes):
    vals = [1]
    for p in primes:
        vals += [v * p for v in vals]
    return set(vals)

# 1. 16-bit vacuum mask
MASK = 57600
assert MASK == 0xE100
assert popcount(MASK) == 4
assert popcount(0xFFFF ^ MASK) == 12

# 2. Squareclass compiler for K = Q(sqrt2, sqrt3, sqrt5)
ALLOWED_SF = {1, 2, 3, 5, 6, 10, 15, 30}
assert squarefree_part(8) == 2
assert squarefree_part(12) == 3
assert squarefree_part(13) == 13
assert squarefree_part(13) not in ALLOWED_SF

# 2b. Wall semantics: cascade wall, not squareclass ceiling
assert 15 in ALLOWED_SF
assert 30 in ALLOWED_SF
assert squarefree_part(15) == 15
assert squarefree_part(30) == 30
assert squarefree_part(13) not in ALLOWED_SF

# 3. Rank-three maximal-constructibility derivation of K
cascade_sf = [squarefree_part(x) for x in [1, 2, 3, 5, 8, 13]]
assert cascade_sf == [1, 2, 3, 5, 2, 13]
assert all(sf in ALLOWED_SF for sf in cascade_sf[:5])
assert cascade_sf[5] not in ALLOWED_SF
assert span_set([2, 3, 5]) == ALLOWED_SF
assert 5 not in span_set([2, 3, 7])
assert 3 not in span_set([2, 5, 7])
assert 2 not in span_set([3, 5, 7])

# 3b. Exhaustive optimality check among small prime triples
primes_for_audit = [2, 3, 5, 7, 11, 13]
passing_first_five = []
for trip in combinations(primes_for_audit, 3):
    S = span_set(list(trip))
    if all(sf in S for sf in cascade_sf[:5]):
        passing_first_five.append(trip)
assert passing_first_five == [(2, 3, 5)]

for trip in combinations(primes_for_audit, 3):
```

```
    S = span_set(list(trip))
    assert not all(sf in S for sf in cascade_sf)

# 4. Palindromic capacity law, closures, 86, 172
C = [2 * ((min(n, 11 - n) + 3) // 2) ** 2 for n in range(12)]
Sigma = []
s = 0
for c in C:
    s += c
    Sigma.append(s)

assert C == C[::-1]
assert C == [2, 8, 8, 18, 18, 32, 32, 18, 18, 8, 8, 2]
assert sum(C[:6]) == 86
assert sum(C) == 172
assert Sigma == [2, 10, 18, 36, 54, 86, 118, 136, 154, 162, 170, 172]

# 4b. Binary capacity and subshell laws
for q, expected in zip([1, 2, 3, 4], [2, 8, 18, 32]):
    assert ((q * q) << 1) == expected

for l, expected in enumerate([2, 6, 10, 14]):
    assert ((l << 2) | 2) == expected

# 5. Facing offsets and facing map
offsets = []
for n in range(12):
    prev = Sigma[n - 1] if n > 0 else 0
    offsets.append(172 - Sigma[n] - prev)

assert offsets == [170, 160, 144, 118, 82, 32, -32, -82, -118, -144, -160, -170]

def floor_of(Z):
    for n in range(12):
        if Z <= Sigma[n]:
            return n
    raise ValueError

def F(Z):
    n = floor_of(Z)
    prev = Sigma[n - 1] if n > 0 else 0
    return Z + 172 - Sigma[n] - prev

for Z in range(1, 173):
    assert 1 <= F(Z) <= 172
    assert F(F(Z)) == Z
    assert F(Z) != Z
    assert F(Z) % 2 == Z % 2

    W = (Z << 8) | F(Z)
    assert byteswap16(W) == ((F(Z) << 8) | Z)
    assert byteswap16(W) == ((F(Z) << 8) | F(F(Z)))
```

```
assert F(86) == 118
assert F(87) == 55
assert ((86 << 8) | F(86)) == 0x5676
assert byteswap16(0x5676) == 0x7656
assert ((118 << 8) | F(118)) == 0x7656

# 6. Exact 2-adic TFI and equatorial diode
def tfi_alpha(Z):
    return popcount(Z ^ (Z - 2))

for Z in range(2, 250):
    expected = nu2(Z) if Z % 2 == 0 else nu2(Z - 1)
    assert tfi_alpha(Z) == expected

assert tfi_alpha(88) == 3
assert tfi_alpha(87) == 1

# 7. Central edge and Kummer wall exposure
A = 86
P = 87
assert A == 0x56
assert P == 0x57
assert A ^ P == 1
assert A + P == 173
assert A + P == 0xAD
assert 173 == 172 + 1
assert (172 & 0x0F) == 12
assert (173 & 0x0F) == 13
assert squarefree_part(12) in ALLOWED_SF
assert squarefree_part(13) not in ALLOWED_SF

# 8. Central FCC cell and Lichtenberg compression
cell = [85, 86, 87, 88]
assert sum(cell) == 346
assert 346 == 2 * 173
assert 346 == 0x15A

def L(n):
    return 2 ** n // 3

for n in range(1, 31):
    assert L(n + 2) == 2 ** n + L(n)

assert L(6) == 21
assert L(8) == 85
assert 43 == 2 * L(6) + 1
assert 85 == 4 * L(6) + 1
assert 2 ** 8 == 3 * L(8) + 1

# 9. Optical phase bit: local LSB inversion
assert P % 2 == 1
assert A % 2 == 0
assert (P ^ 1) == A
```

```
assert (A ^ 1) == P

# 10. 137 register identities and cascade shadows
assert 137 == 0x89
assert 137 == 13 * 13 - 32
assert 137 == 240 - 103
assert popcount(137) == 3
assert 1 * 2 * 3 * 5 * 8 == 240
assert 1 + 2 + 3 + 5 + 8 + 13 == 32
assert 1 + 2 + 3 + 5 + 8 + 13 + 21 + 34 == 87
assert 12 * 12 == 144
assert 1728 == 12 ** 3

# 11. FCC closure shifts
assert 86 << 1 == 172
assert 43 << 2 == 172
assert 172 == 4 * 43

# 12. Heegner and Ogg boundary audits
heegner_primes = [2, 3, 7, 11, 19, 43, 67, 163]
monster_primes = [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 41, 47, 59, 71]

assert [p for p in heegner_primes if p in monster_primes] == [2, 3, 7, 11, 19]
assert min(p for p in heegner_primes if p not in monster_primes) == 43
assert all(p % 24 == 19 for p in heegner_primes if p > 11)
assert 43 - 19 == 24
assert 67 - 43 == 24
assert 163 - 67 == 96
assert 960 == 4 * 240
assert 960 ** 3 == 884736000

# 13. Tetrahedral angle witness
angle = math.degrees(math.acos(-1.0 / 3.0))
assert abs(angle - 109.471220634) < 1e-6

# 14. H2 Bond Scale Verification (first numerical bridge)
Ry = 13.605693122994
T_min = Ry / 3.0
D0_H2 = 4.478
assert abs(T_min - 4.535231040998) < 1e-9
assert abs((T_min - D0_H2) / D0_H2) < 0.02  # ~1.3% error

# 15. Spectral quadratic form witness
def qform(n1, n2, n3):
    return n1 * n1 + n2 * n2 / 2.0 + n3 * n3 / 3.0

assert abs(qform(0, 0, 1) - 1.0 / 3.0) < 1e-15
assert abs(qform(1, 0, 0) - 1.0) < 1e-15
assert abs(qform(0, 1, 0) - 0.5) < 1e-15

print("All extended Level A, register, optimality, wall-semantics, H2-bridge, Heegner/Ogg, and
```

E The First Dynamical Bridge: Toroidal Spectrum and the H_2 Bond Scale

This appendix provides the first explicit numerical derivation linking the abstract IT^3 topology to measurable physical units in electron-volts. It elevates the model from a structural architecture to a predictive framework with a falsifiable number.

Theorem E.1 (Additive toroidal spectrum)**.** *Let*

$$T^3 = \mathbb{R}^3/\Lambda$$

be a flat three-torus with orthogonal periods L_1, L_2, L_3. In a trivial local curvature sector, the Fourier modes

$$\psi_{\mathbf{n}}(x) = \exp\left(2\pi i \sum_{j=1}^{3} \frac{n_j x_j}{L_j}\right), \qquad \mathbf{n} \in \mathbb{Z}^3,$$

satisfy

$$D^2\psi_{\mathbf{n}} = (2\pi)^2 \left(\frac{n_1^2}{L_1^2} + \frac{n_2^2}{L_2^2} + \frac{n_3^2}{L_3^2}\right)\psi_{\mathbf{n}}.$$

Hence orthogonal metric contributions add quadratically.

Proof. On a flat torus the spin connection may be trivialized in the chosen sector. The Dirac operator reduces to the first-order Fourier symbol. Squaring gives the dual-lattice quadratic form. Orthogonality of the periods makes cross terms vanish. □

Corollary E.2 (Pythagorean composition is spectral)**.** *The additive composition of squared metric tensions is the arithmetic shadow of the Laplacian/Dirac spectrum on T^3. It does not need to be postulated separately once the flat torus spectral model is accepted.*

Now impose the rank-three constructible metric selected in Section 3. With one overall length scale a, the spatial periods are

$$L_1 = a, \qquad L_2 = a\sqrt{2}, \qquad L_3 = a\sqrt{3}.$$

Then

$$D^2\psi_{\mathbf{n}} = \frac{(2\pi)^2}{a^2}\left(n_1^2 + \frac{n_2^2}{2} + \frac{n_3^2}{3}\right)\psi_{\mathbf{n}}.$$

The lowest non-scalar spatial squareclass denominator is therefore 3.

Conditional Theorem E.3 (The Toroidal Kinetic Quantum)**.** In atomic units ($\hbar = m_e = e = 1$), the Rydberg energy

$$\mathrm{Ry} = 13.605693122994\ldots \text{ eV}$$

establishes the fundamental chemical energy scale. For a wave localized on a torus of period L (in units of the Bohr radius a_0), the minimum kinetic energy of the mode is:

$$T(L) = \frac{\mathrm{Ry}}{L^2}.$$

Theorem E.4 (Derivation of the H_2 Bond Energy Scale)**.** *The leading stabilization energy of the covalent bond in H_2 is structurally governed by the longest spatial root of the K field. The lowest non-trivial spatial mode corresponds to the $\sqrt{3}$ axis, giving*

$$T_{\min} = \frac{\mathrm{Ry}}{(\sqrt{3})^2} = \frac{\mathrm{Ry}}{3} \approx 4.535 \text{ eV}.$$

The experimental dissociation energy is $D_0(H_2) \approx 4.478$ eV. The relative deviation is

$$\delta = \frac{|4.535 - 4.478|}{4.478} \approx 1.3\%.$$

Proof. By Theorem E.1 and the rank-three metric $1, \sqrt{2}, \sqrt{3}$, the eigenvalues of D^2 are

$$\lambda_{\vec{n}}^2 \propto n_1^2 + \frac{n_2^2}{2} + \frac{n_3^2}{3}.$$

The longest wavelength (lowest kinetic energy) corresponds to the largest period, i.e. the $\sqrt{3}$ axis with $n_3 = 1$, $n_1 = n_2 = 0$:

$$\lambda_{\min}^2 \propto \frac{1}{3}.$$

Converting to energy via the Rydberg scale:

$$T_{\min} = \frac{\mathrm{Ry}}{3} = \frac{13.605693}{3} = 4.535231\ldots \text{ eV}.$$

The H_2 molecule is a pure s-s interference bond occupying the longest available spatial mode. The agreement to within 1.3% is obtained without any free parameter fitted to H_2. □

Object	Register input	Bridge scale	Experiment	Status
H_2 bond	squareclass denom. 3	Ry /3 = 4.535 eV	$D_0 \approx 4.478$ eV	Level B

Numerology Guard E.5 (Status of the H_2 bridge). This is not a derivation of the full molecular bond energy. It is a falsifiable scale diagnostic. The arithmetic value Ry /3 is Level A. The identification of this value with the leading H_2 bonding scale is Level B, conditional on the toroidal-hopping calibration. If accurate quantum-chemical decompressions show that no leading bonding scale of order Ry /3 exists, this bridge fails.

Proposition E.6 (Qualitative He_2 cancellation)**.** *In the two-mode register architecture, the bonding and antibonding channels are separated by one spinor bit. For* H_2*, one spinor pair occupies the bonding channel. For* He_2*, the next pair must occupy the antibonding channel. At leading order:*

$$2\beta + 2(-\beta) = 0,$$

while nuclear repulsion remains. Therefore He_2 *is not predicted to possess an ordinary covalent ground-state bond.*

Proof. Standard two-level cancellation written in register language. The register supplies only one spinor bit per spatial channel, so the pair structure is rigid. □

Numerology Guard E.7. The He_2 statement is qualitative Level B. It does not compute the van der Waals bond of He_2, which is a correlation effect outside the leading two-mode register model.

Research Target E.8 (Continuum limit and emergent isotropy)**.** Prove that when the torus scale $a \to 0$, the algebraic wall

$$\Lambda_{\text{wall}} \sim \frac{\sqrt{13}}{a}$$

moves to the ultraviolet, while the low-energy chemistry reproduces ordinary isotropic quantum mechanics. The anisotropic metric ratio $1 : \sqrt{2} : \sqrt{3}$ must produce only corrections below chemical precision at eV scales. This is Level O.

F Falsification Protocol

To strictly distinguish the IT^3 framework from post-hoc numerology, the following explicit bets are recorded. Each one can be wrong.

Research Target F.1 (Falsifier 1: The Absolute Ceiling at $Z = 172$)**. Prediction:** The element $Z = 173$ is physically impossible as a bound neutral atom. The source node $173 = \texttt{0xAD}$ possesses an illegal low nibble $\texttt{D} = 13$. The 16-bit register cannot map this state.

Refutation: If a neutral atom with $Z \geq 173$ is synthesized and exhibits ordinary periodic chemical structure, the 16-bit FCC mask topology is falsified.

Research Target F.2 (Falsifier 2: Hydrogen Bridge)**. Prediction:** The leading single-bond stabilization scale is

$$E_{\text{bridge}} = \frac{\text{Ry}}{3} \approx 4.535 \text{ eV}.$$

Refutation: If accurate quantum-chemical decompressions show that H_2 bonding is not controlled at leading order by a dimensionless factor of order 1/3 of the Rydberg scale, this bridge fails.

Research Target F.3 (Falsifier 3: 2-adic Kinetic Penalty in α-Decay)**.** The Topological Friction Index dictates that an α-decay crossing the Relativistic Equator experiences different binary borrow cascades depending on its parity:

- Even rail ($88 \to 86$): TFI = 3.
- Odd rail ($87 \to 85$): TFI = 1.

Prediction: After correcting for Geiger–Nuttall systematics and angular-momentum hindrance, a large alpha-decay dataset should show a residual even/odd asymmetry at closures $E \equiv 6 \pmod 8$.

Refutation: If no such residual asymmetry exists, the diode remains binary arithmetic, not nuclear physics.

Research Target F.4 (Falsifier 4: Technetium Demotion)**.** Unless a quantitative nuclear instability is derived from the spectral action, $Z = 43$ is only an Ogg/Heegner resonance (Level C). It must not be used as a predictive explanation of technetium radioactivity.

Refutation: This is self-imposed. If the framework continues to claim Technetium instability as a prediction without a dynamical derivation, it violates its own epistemic protocol.

Research Target F.5 (Falsifier 5: Continuum Compatibility)**.** If the algebraic wall $\sqrt{13} \notin K$ produces visible violations of ordinary low-energy quantum chemistry at eV scales, the model is dead. A successful theory must contain the limit

$$T^3 \to \mathbb{R}^3, \qquad \Lambda_{\text{wall}} \to \infty,$$

in which standard QED and Hartree–Fock chemistry are recovered to known precision.

Refutation: If the anisotropic metric $1 : \sqrt{2} : \sqrt{3}$ leaves detectable fingerprints in atomic spectra or molecular geometries beyond experimental uncertainty, the toroidal model is falsified at low energy.

G The Dynamical Frontier: Deriving the Postulates from Spectral Geometry

The present paper establishes that if the vacuum obeys four geometric/architectural postulates (toroidal rank 3, orthogonal metric composition, Pythagorean–Fibonacci growth, and maximal constructibility delay), then the field

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}),$$

the Kummer wall $\sqrt{13} \notin K$, the 16-bit covolume mask $\texttt{0xE100}$, the FCC closure $172 = 43 \ll 2$, the Ogg boundary, and the chemical register architecture follow as exact arithmetic or conditional theorems.

The next frontier of the IT^3 framework is to eliminate these postulates by deriving them as strict dynamical minima of a fundamental action. In the context of Noncommutative Geometry (NCG), the natural candidate is the Chamseddine–Connes spectral action [1, 4]:

$$S_{\text{spec}} = \text{Tr}\left(f\left(\frac{D_A}{\Lambda}\right)\right) + \langle\psi, D_A\psi\rangle + S_{\text{ferm}},$$

where D_A is the gauged Dirac operator of a real spectral triple, Λ is the ultraviolet cutoff scale, and f is a positive even function.

For the present arena

$$M_{\text{tot}} = \mathbb{S}^4 \times \mathbb{T}^3,$$

the spectral action is the natural dynamical object because it simultaneously sees the topology of the compact space, the holonomy of the flat torus fibre, and the discrete arithmetic of the constructible field K.

The following four targets are therefore not philosophical remarks. They are precise Level O problems: each one, if solved, upgrades a present Level B postulate to a Level A dynamical theorem.

Epistemic Status G.1 (Level O convention). Nothing in this appendix is used as numerical evidence. Level O statements carry no arithmetical authority by themselves. They define the exact boundary between the present algebraic/topological skeleton and a future dynamical theory.

Research Target G.2 (Dynamical generation of the toroidal rank)**. Problem.** Derive Postulate 3.1

$$\dim_{\mathbb{F}_2} H^1(\mathbb{T}^3; \mathbb{F}_2) = 3$$

from the stability of the vacuum manifold.

Pathway. One must show that the relevant minimum of the spectral action, in a fixed topological sector, forces the internal space to factor as

$$\mathbb{S}^4 \times \mathbb{T}^3,$$

with the flat $\mathbb{T}^3$ absorbing the infrared continuous symmetries and leaving exactly three independent mod-2 one-cycles for the metric squareclass group.

Research Target G.3 (D_A^2-spectrum and orthogonal composition)**. Problem.** Derive Postulate 3.2, namely the additive Pythagorean composition of orthogonal squared metric tensions, from the spectrum of the gauged Dirac operator.

Pathway. The Lichnerowicz–Weitzenböck formula gives

$$D_A^2 = \nabla_A^*\nabla_A + \frac{1}{4}\,\text{Scal} + \frac{1}{2}\,\gamma^{\mu\nu}F_{\mu\nu}.$$

On the flat torus fibre $\mathbb{T}^3$, the scalar curvature vanishes and the topological twist is modeled by a flat $U(1)$ connection with nontrivial holonomy. Locally, $F_{\mu\nu} = 0$, so the spectrum is governed by the dual lattice quadratic form.

Research Target G.4 (Arithmetic entropy and maximal delay)**. Problem.** Derive Postulates 3.3 and 3.4, i.e. the Fibonacci cascade and the maximal-delay principle, from an entropy or variational principle.

Pathway. Let $\Gamma \subset \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ be the squareclass group of allowed metric tensions. Define its arithmetic complexity by

$$c(\Gamma) = \dim_{\mathbb{F}_2} \Gamma.$$

Introducing a new independent prime squareclass increases $c(\Gamma)$ and therefore increases the number of independent algebraic directions available to the vacuum register.

Research Target G.5 (Spectral distance, infinite refractive index, and confined light)**. Problem.** Derive the Confined-Light Ontology, Axiom 10.5, from the spectral distance of noncommutative geometry.

Pathway. In Connes' formulation, the distance between two states is recovered from the Dirac operator by

$$d(x, y) = \sup \{|a(x) - a(y)| \ : \ \|[D, a]\| \leq 1\} .$$

If the allowed coordinate algebra is the constructible algebra over

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}),$$

then a metric direction requiring $\sqrt{13}$ is not representable inside the algebra. As a would-be propagating mode approaches the Kummer wall, the commutator seminorm cannot remain finite while separating the endpoint from the bulk. In the spectral metric this appears as

$$d(x, y) \to \infty.$$

Numerology Guard G.6 (Epistemic status of Appendix G). Appendix G contains no Level A proof of quantum gravity, no derivation of the Standard Model, and no numerical prediction. It is a precise research programme. Its purpose is to separate cleanly what has already been proved arithmetically from what remains to be derived dynamically.

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